

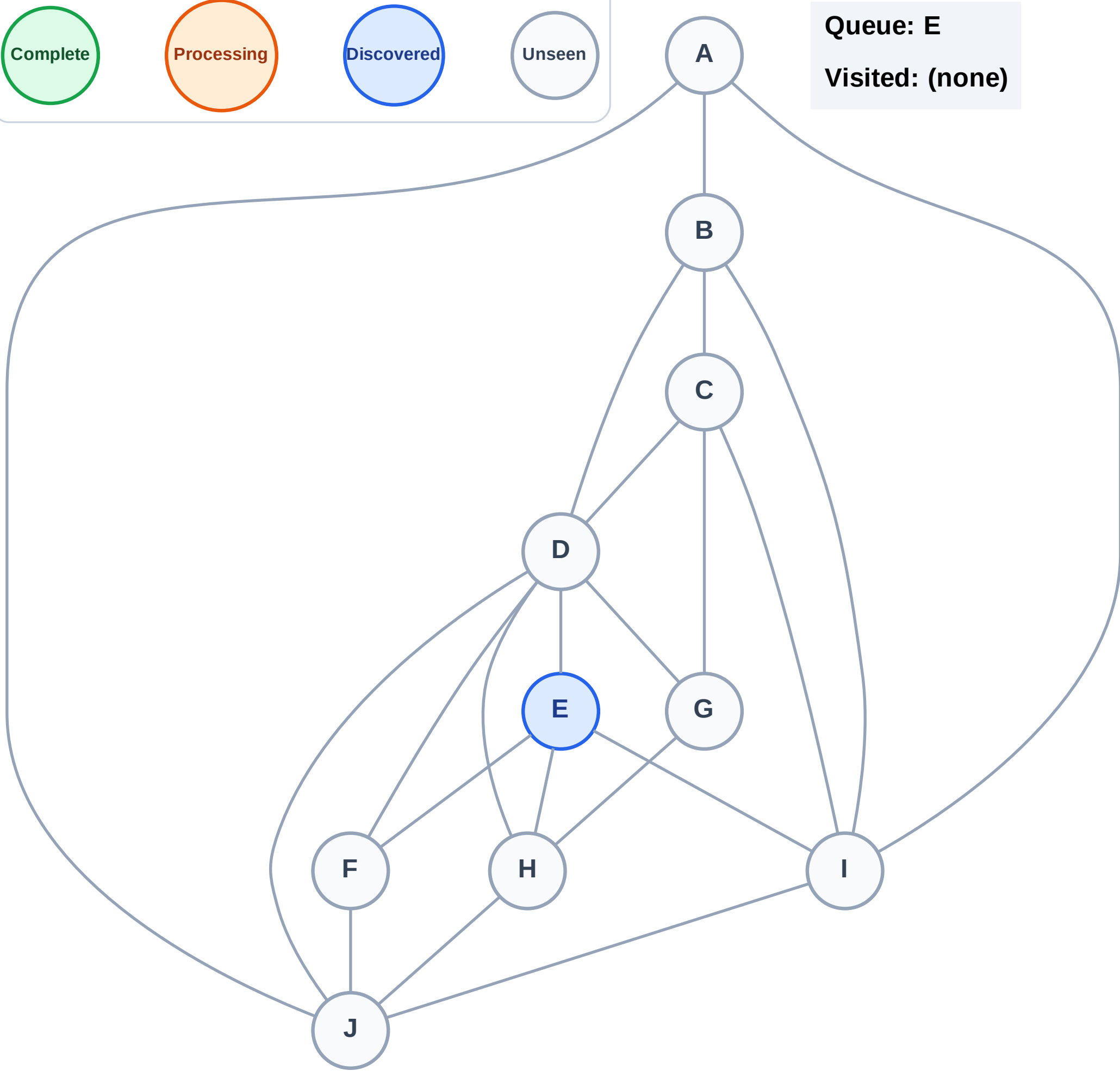
BFS Traversal

Step 1: Start at E

Legend



Queue: E
Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

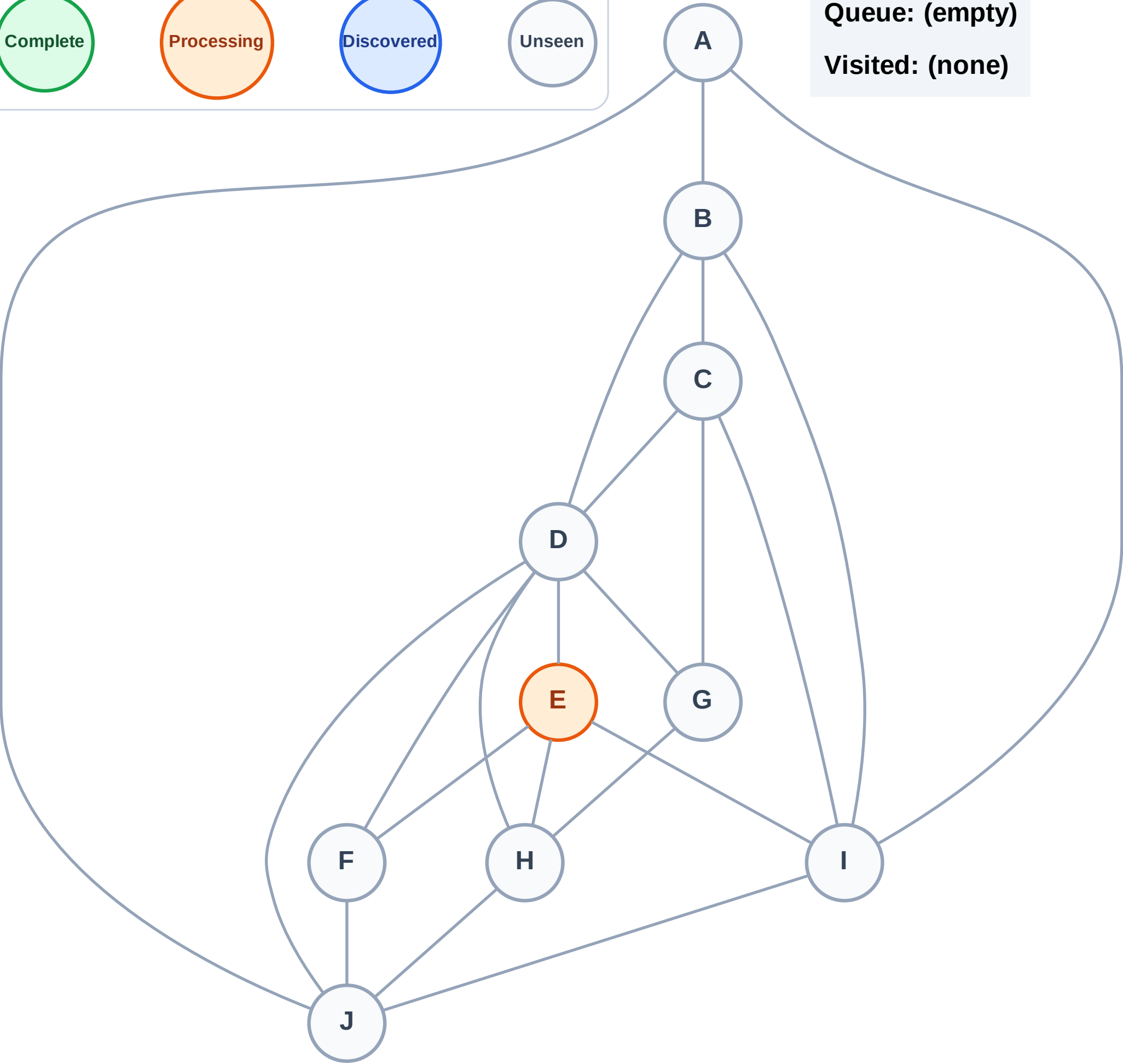
Complete

Processing

Discovered

Unseen

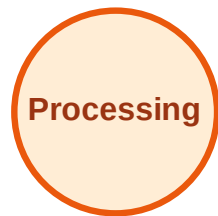
Queue: (empty)
Visited: (none)



BFS Traversal

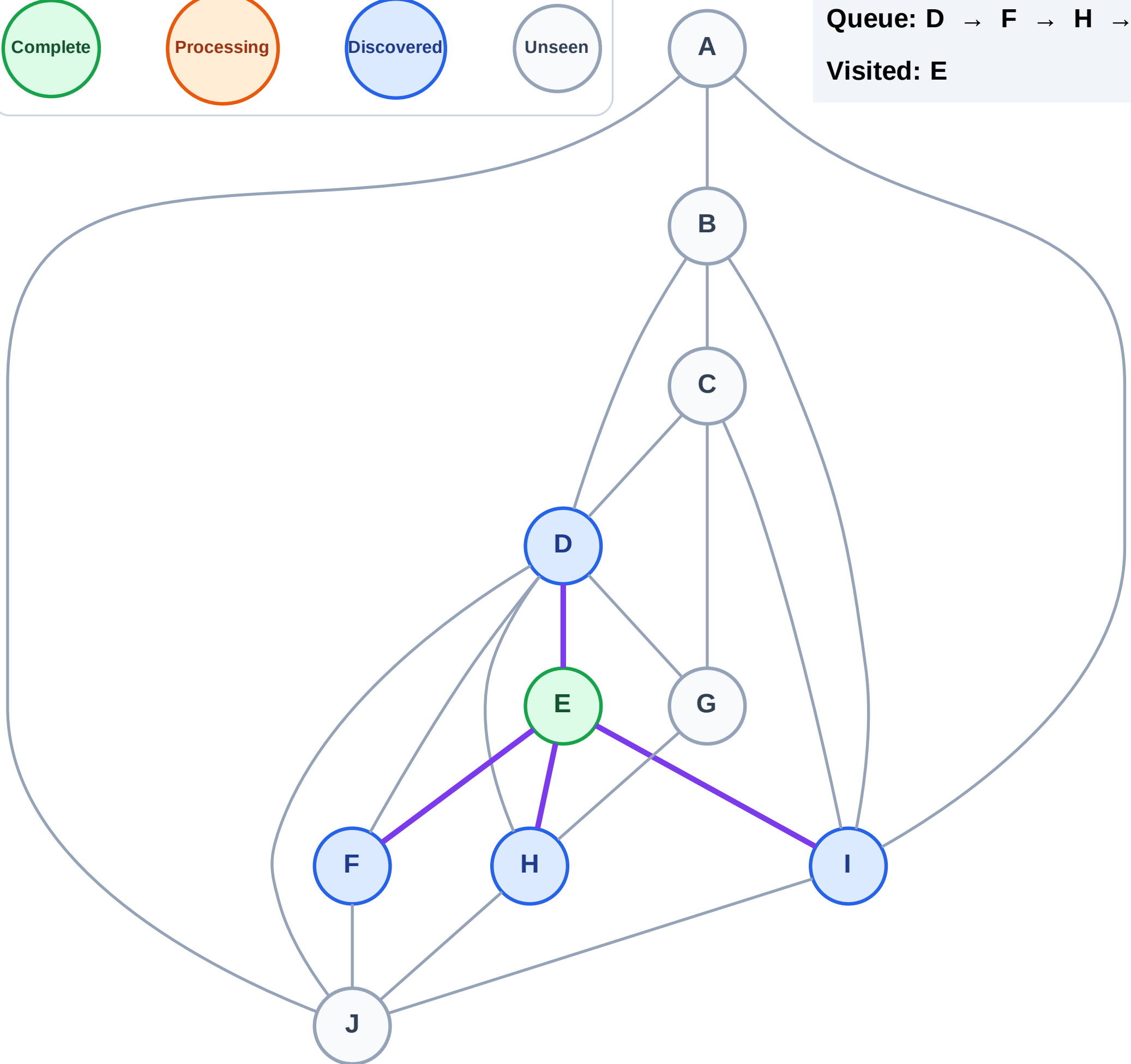
Step 3: Discover D, F, H, I from E

Legend



Queue: D → F → H → I

Visited: E



BFS Traversal

Step 4: Process node D

Legend

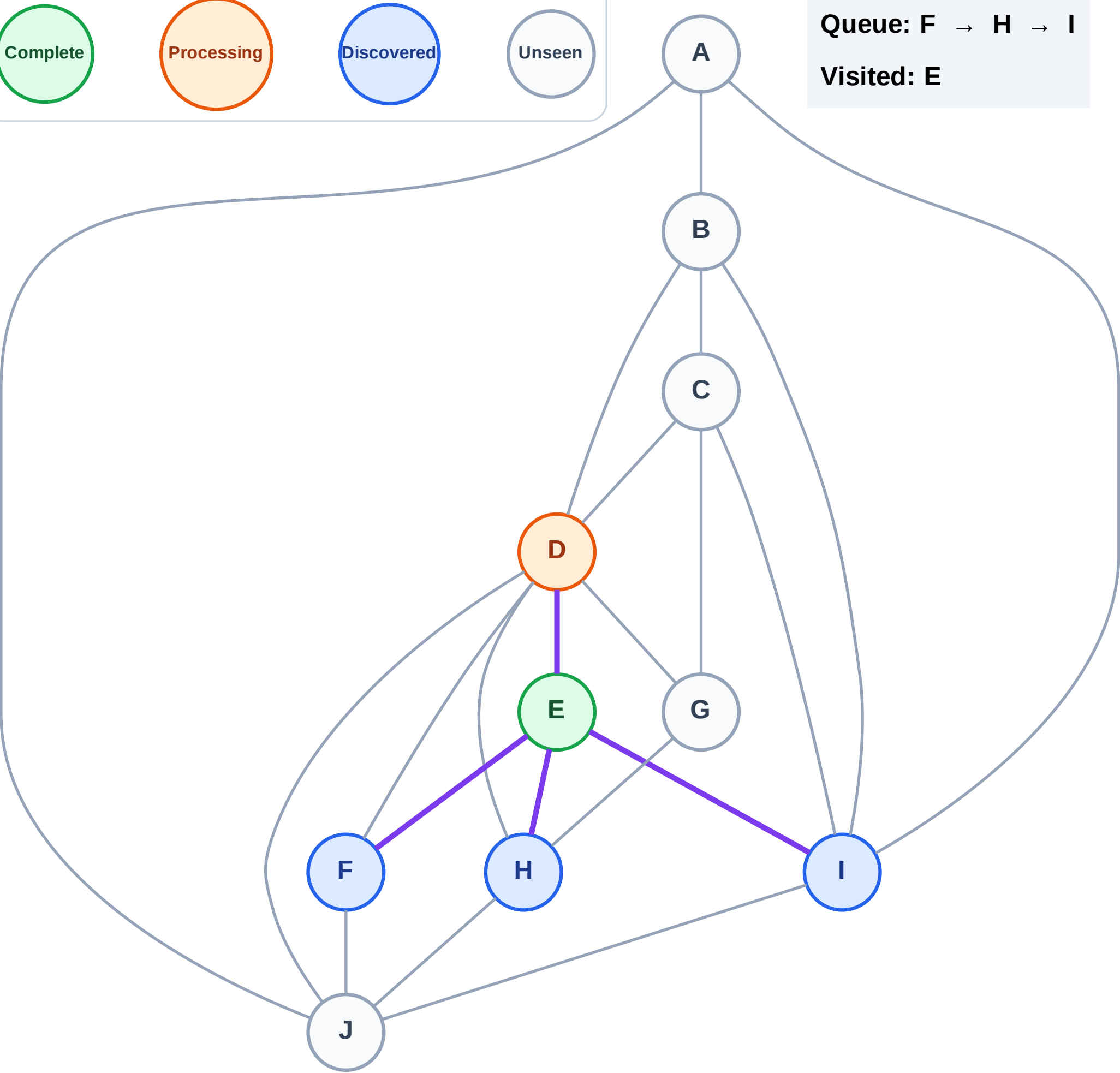
Complete

Processing

Discovered

Unseen

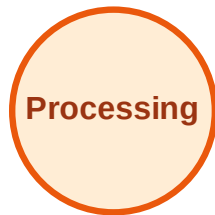
Queue: F → H → I
Visited: E



BFS Traversal

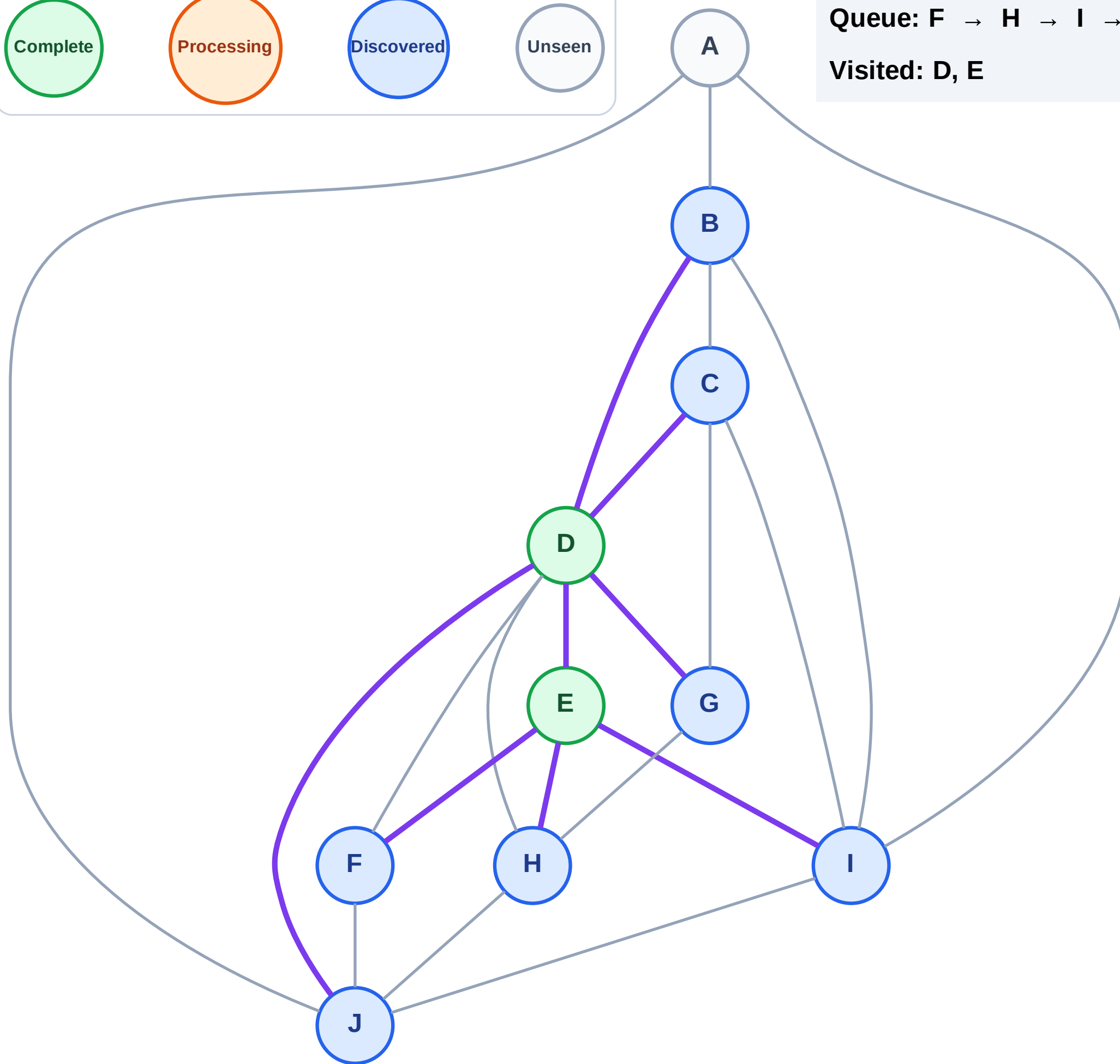
Step 5: Discover B, C, G, J from D

Legend



Queue: F → H → I → B → C → G → J

Visited: D, E



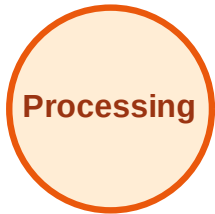
BFS Traversal

Step 6: Process node F

Legend



Complete



Processing



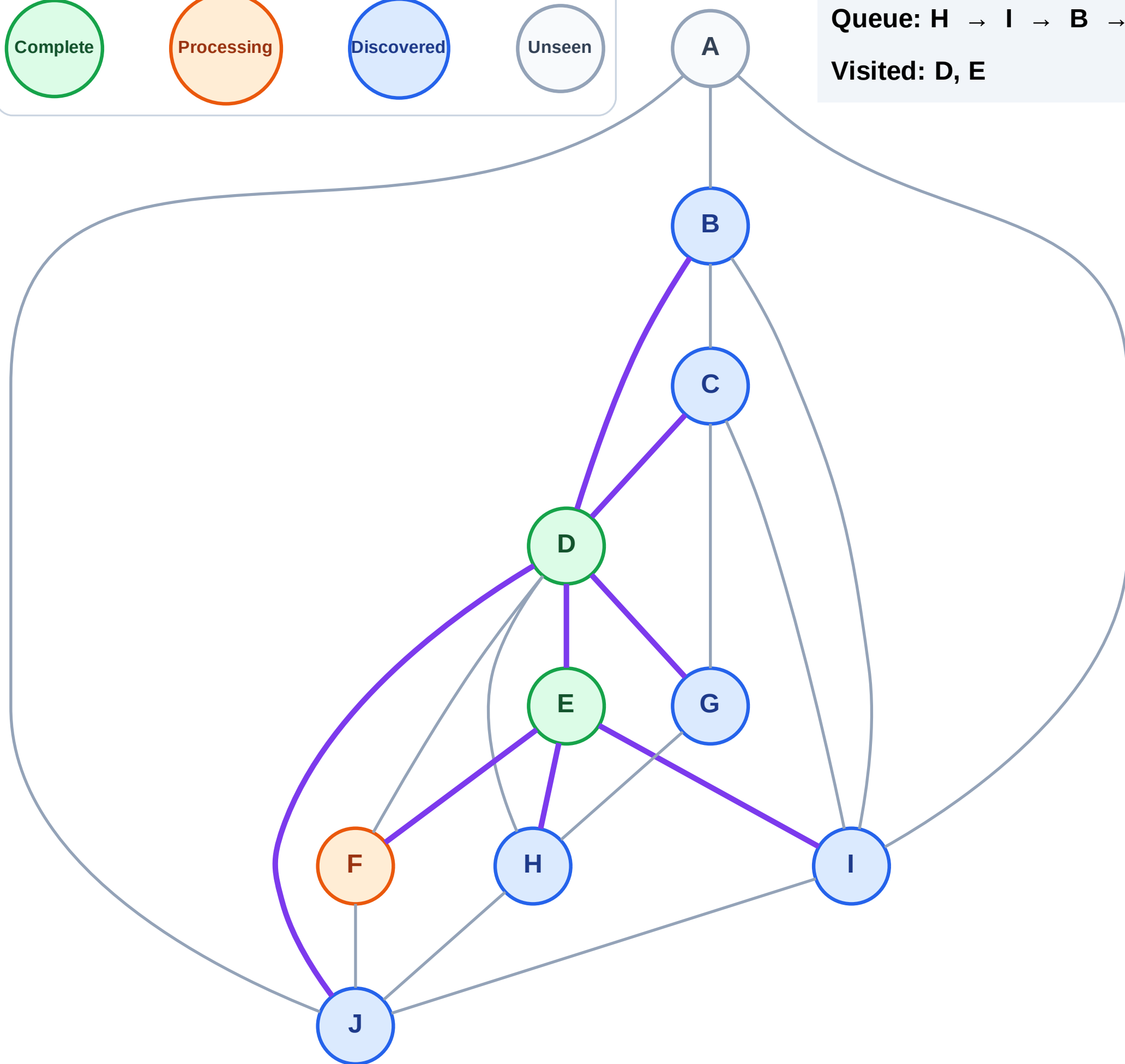
Discovered



Unseen

Queue: H → I → B → C → G → J

Visited: D, E



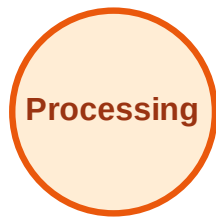
BFS Traversal

Step 7: No new neighbors from F

Legend



Complete



Processing



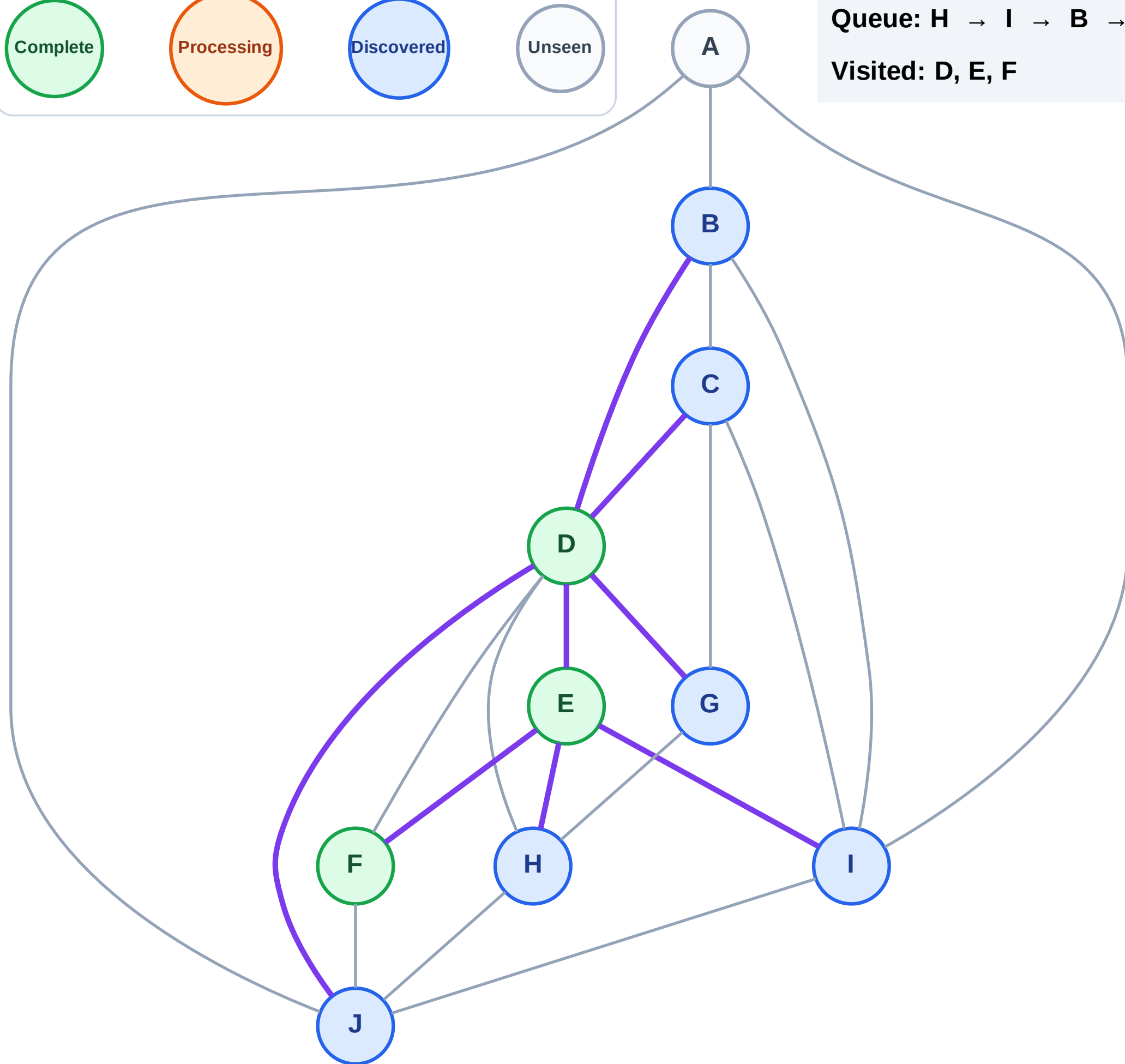
Discovered



Unseen

Queue: H → I → B → C → G → J

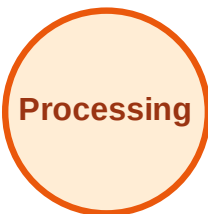
Visited: D, E, F



BFS Traversal

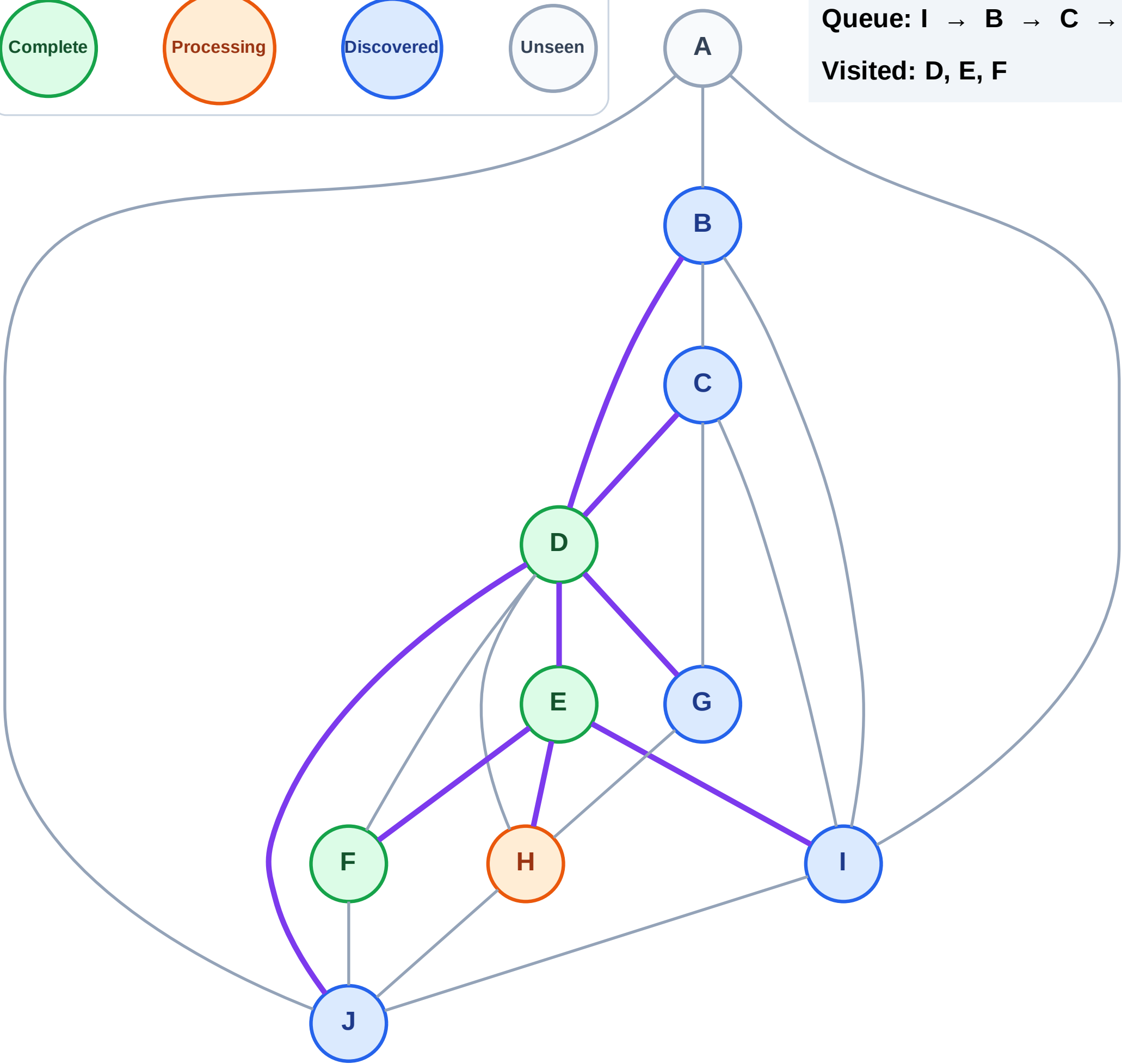
Step 8: Process node H

Legend



Queue: I → B → C → G → J

Visited: D, E, F



BFS Traversal

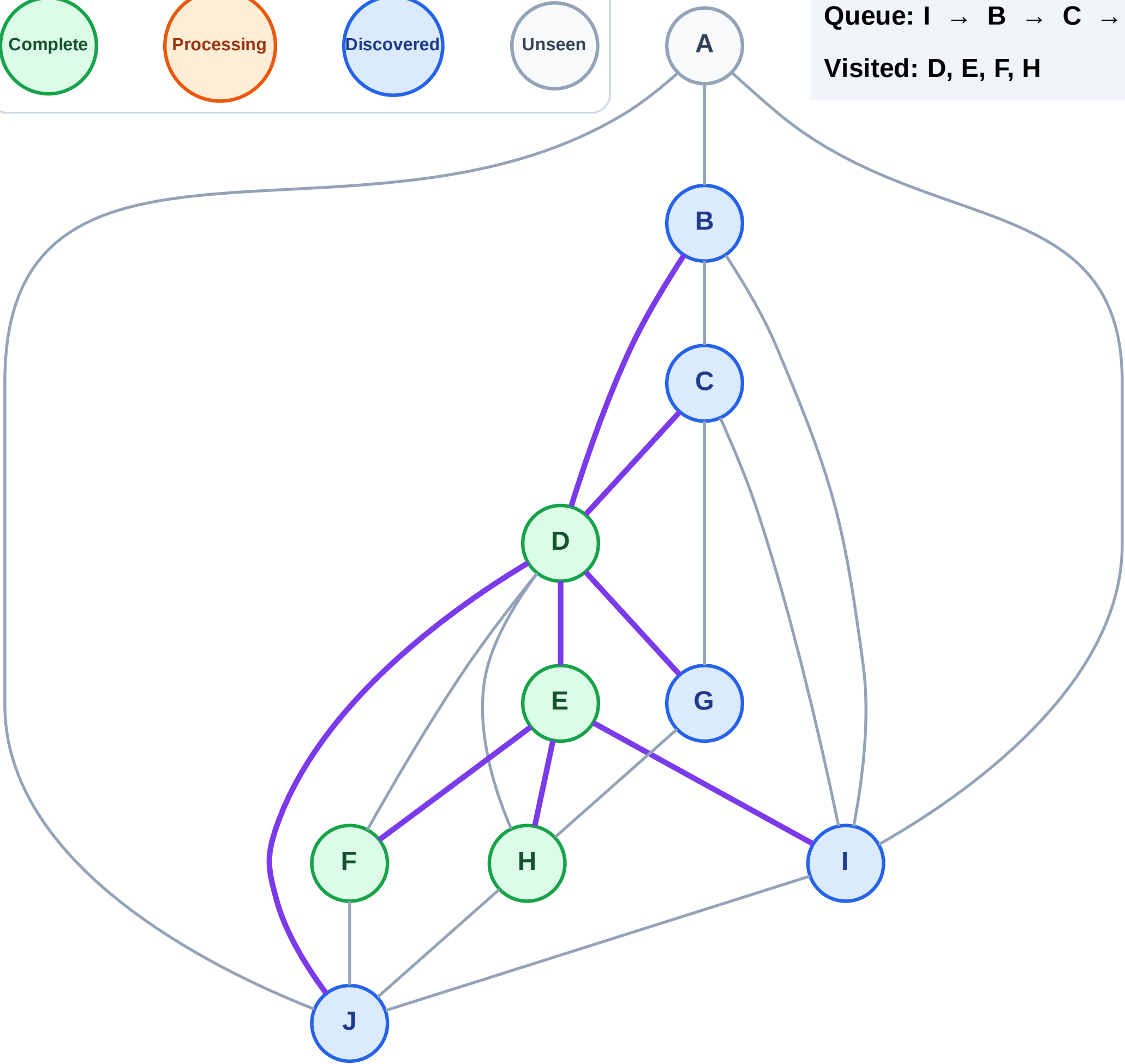
Step 9: No new neighbors from H

Legend



Queue: I → B → C → G → J

Visited: D, E, F, H



BFS Traversal

Step 10: Process node I

Legend

Complete

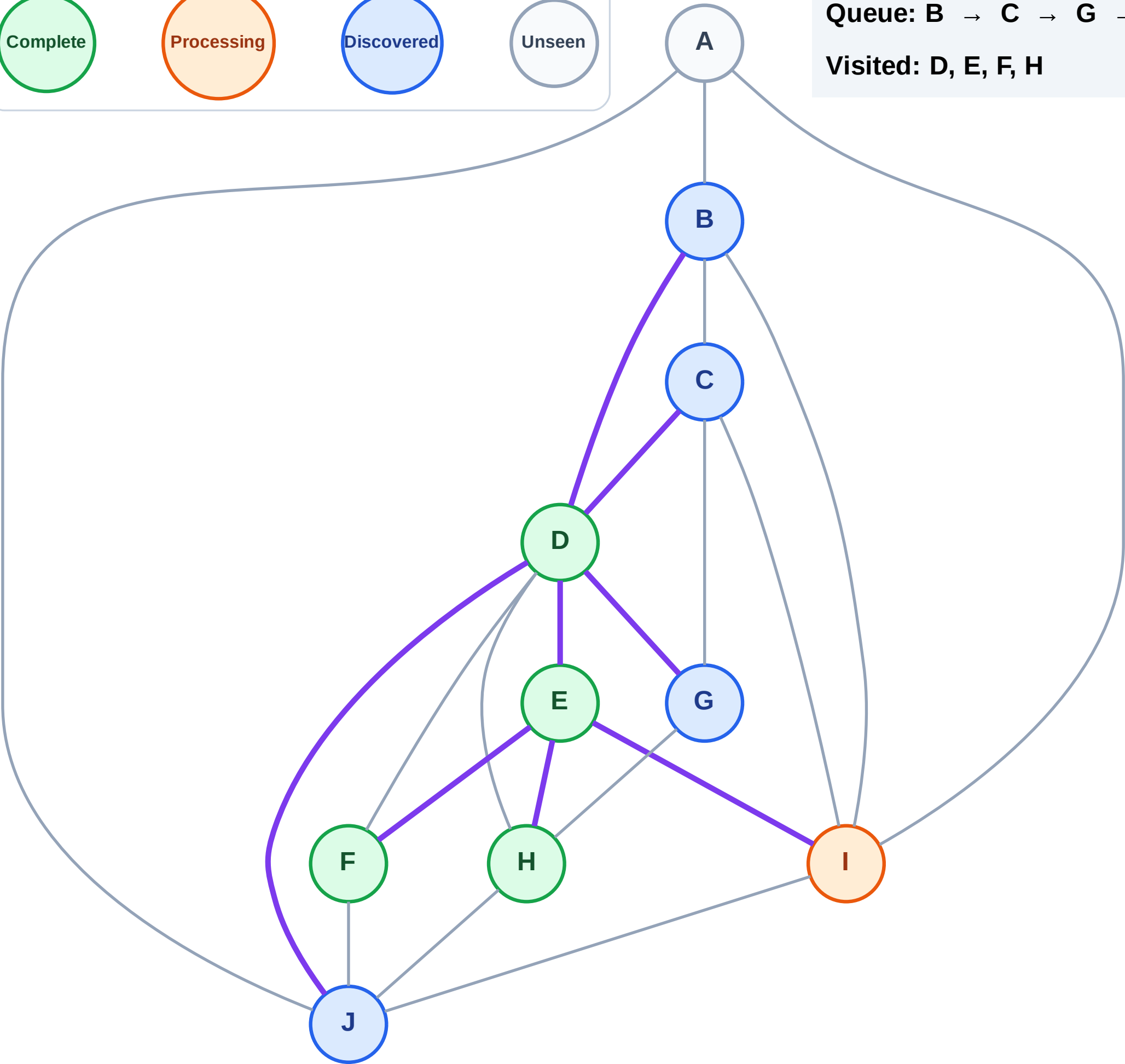
Processing

Discovered

Unseen

Queue: B → C → G → J

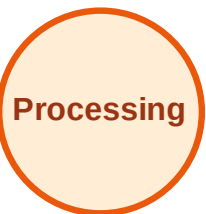
Visited: D, E, F, H



BFS Traversal

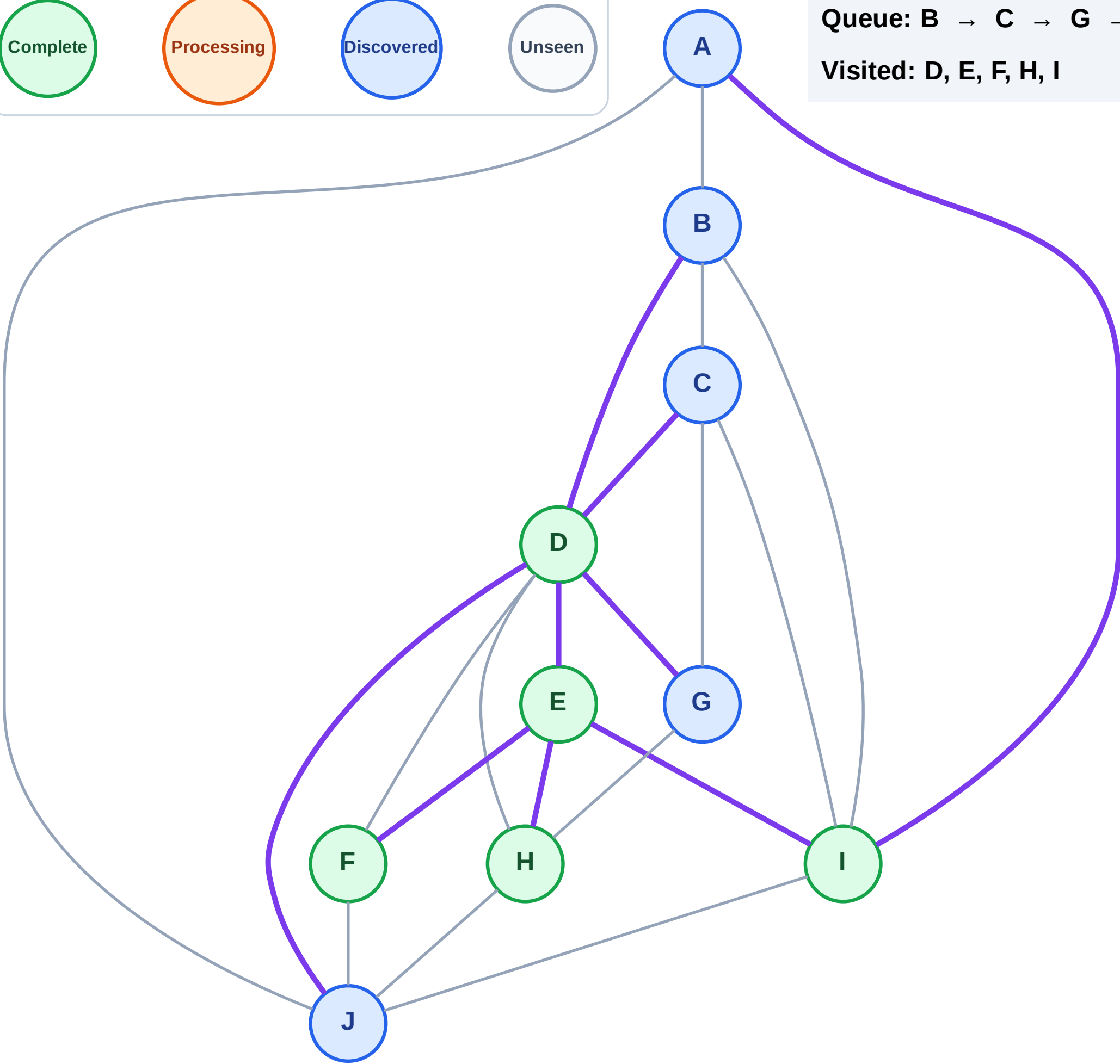
Step 11: Discover A from I

Legend



Queue: B → C → G → J → A

Visited: D, E, F, H, I



BFS Traversal

Step 12: Process node B

Legend

Complete

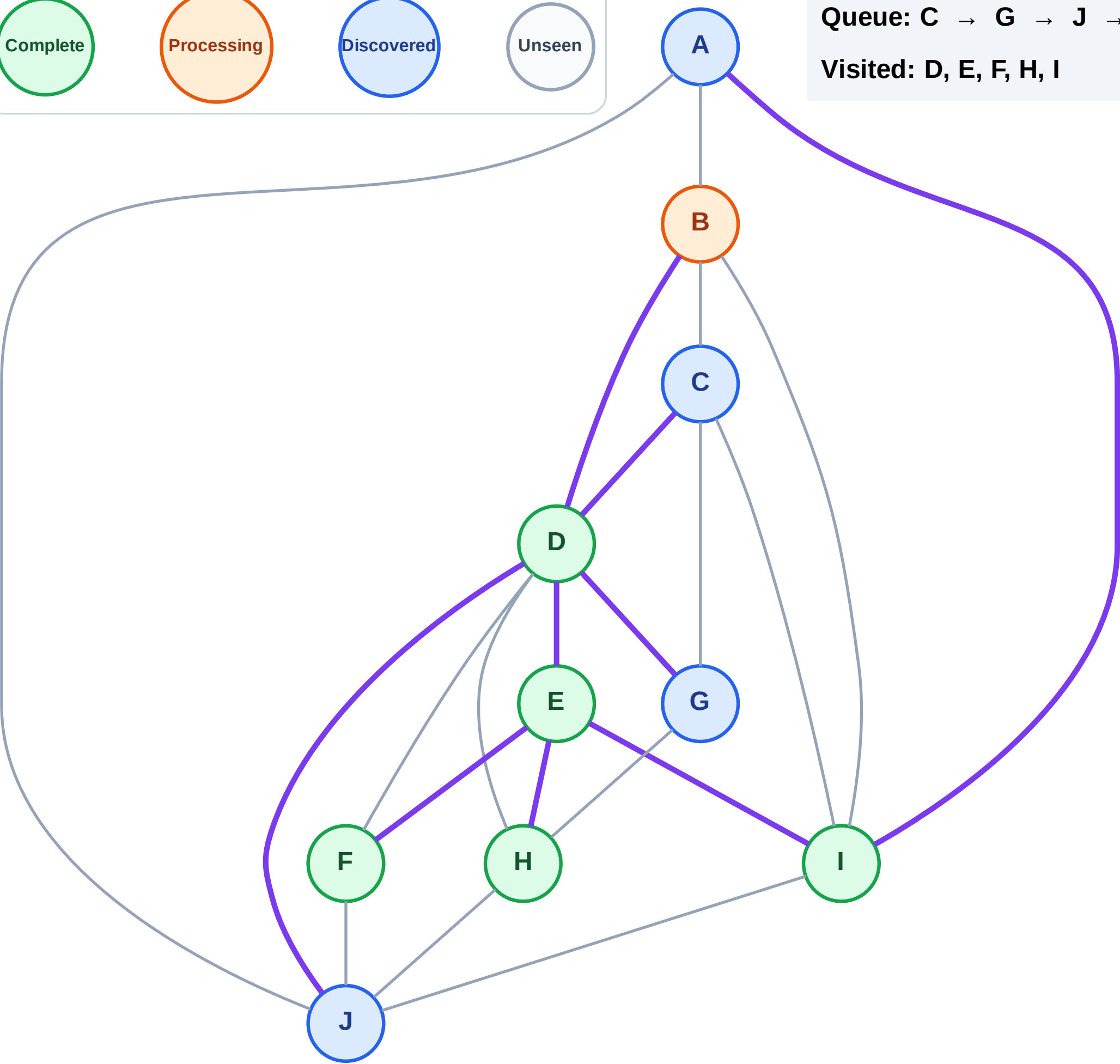
Processing

Discovered

Unseen

Queue: C → G → J → A

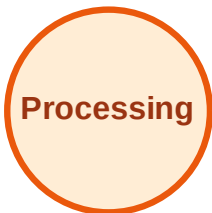
Visited: D, E, F, H, I



BFS Traversal

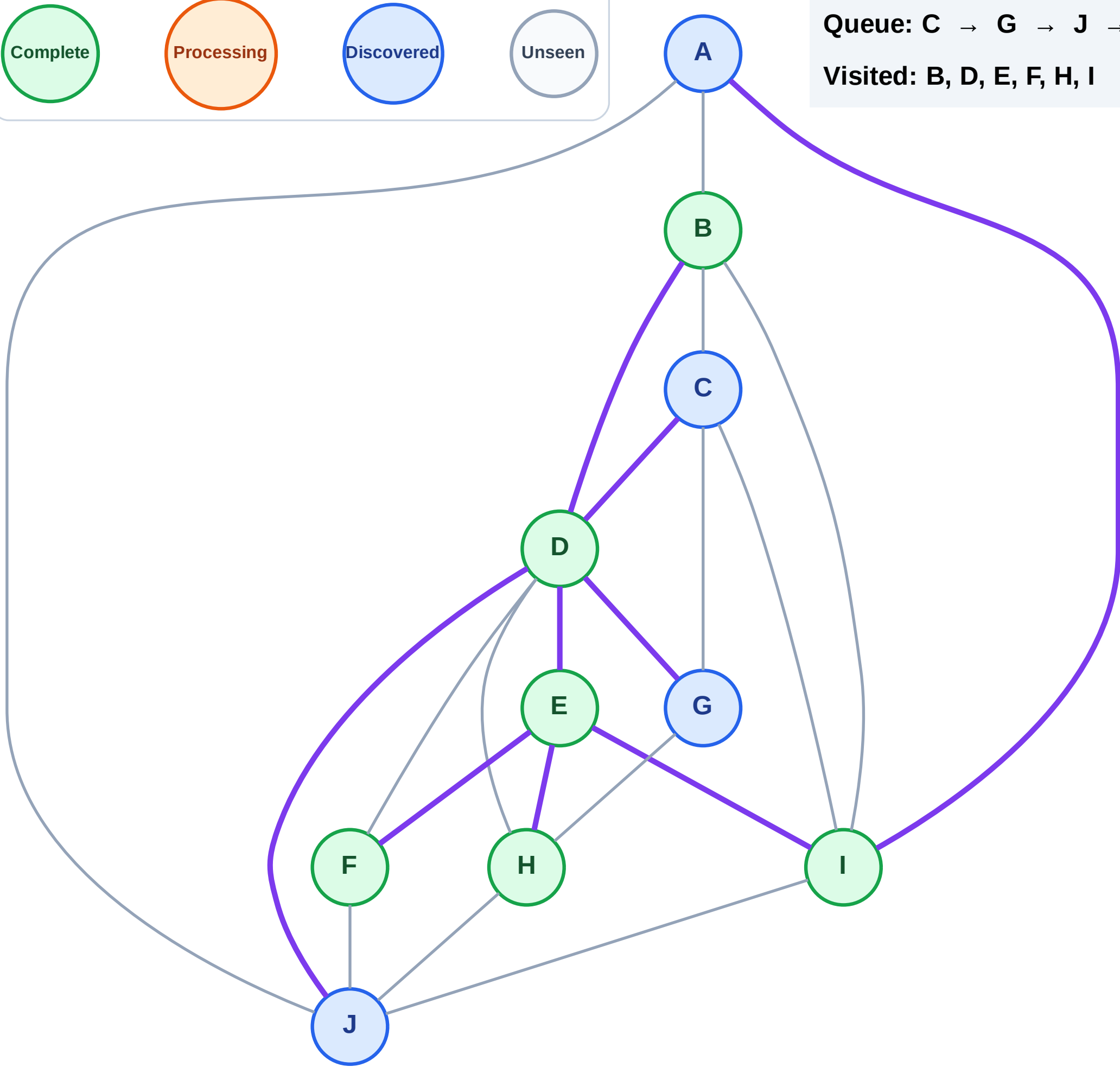
Step 13: No new neighbors from B

Legend



Queue: C → G → J → A

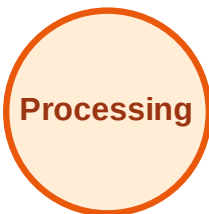
Visited: B, D, E, F, H, I



BFS Traversal

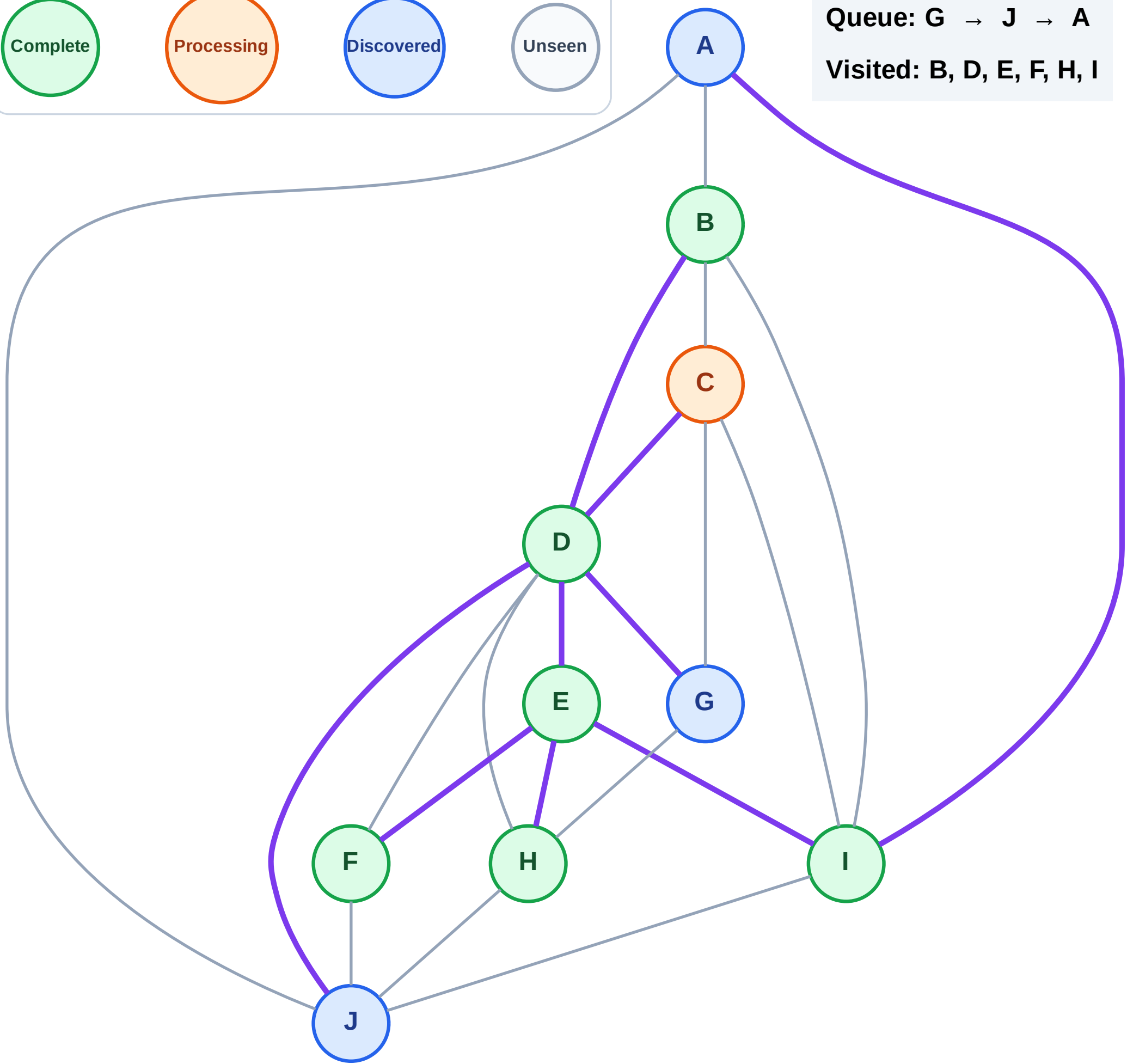
Step 14: Process node C

Legend



Queue: G → J → A

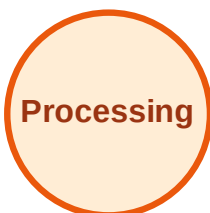
Visited: B, D, E, F, H, I



BFS Traversal

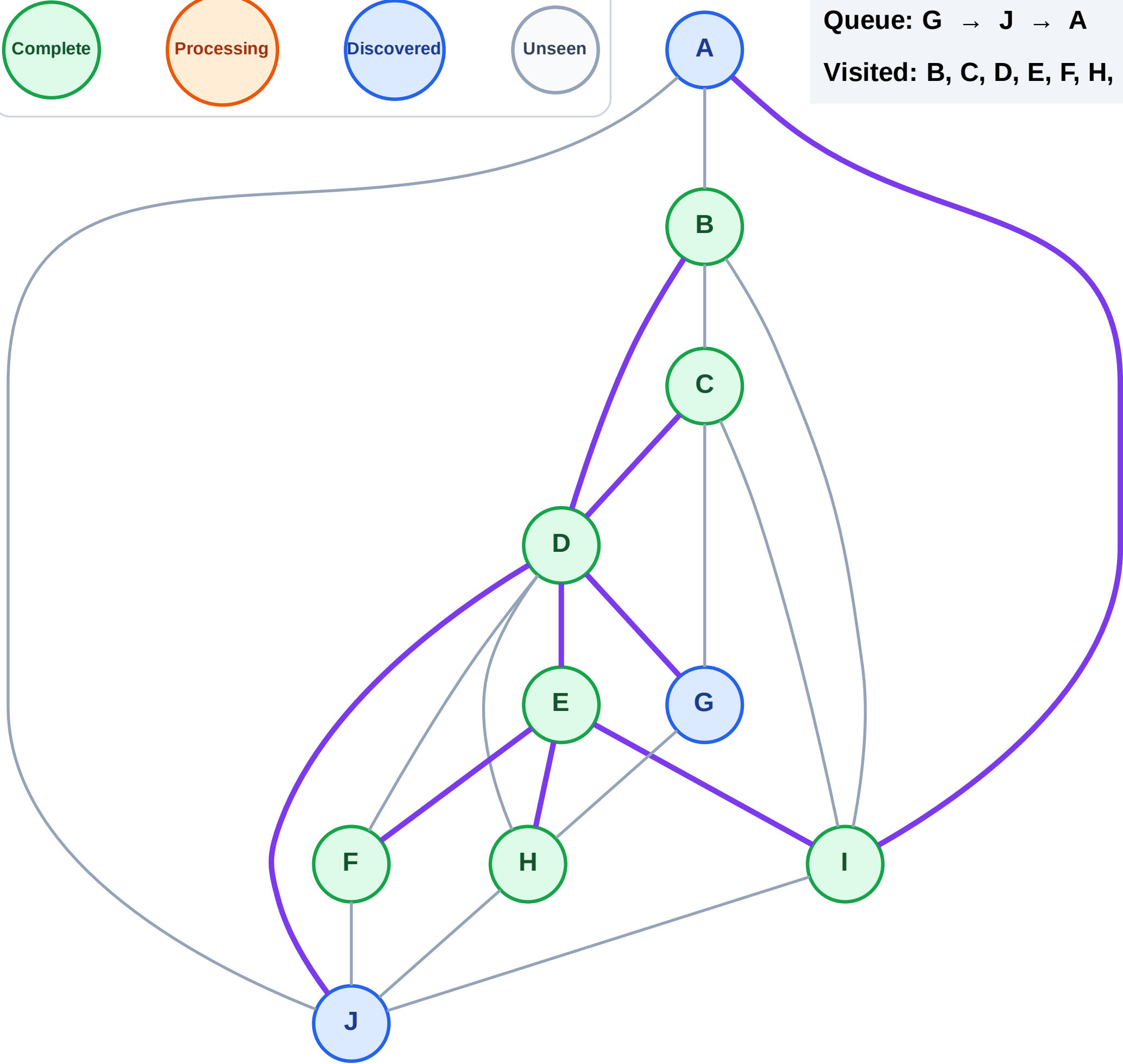
Step 15: No new neighbors from C

Legend



Queue: G → J → A

Visited: B, C, D, E, F, H, I



BFS Traversal

Step 16: Process node G

Legend

Complete

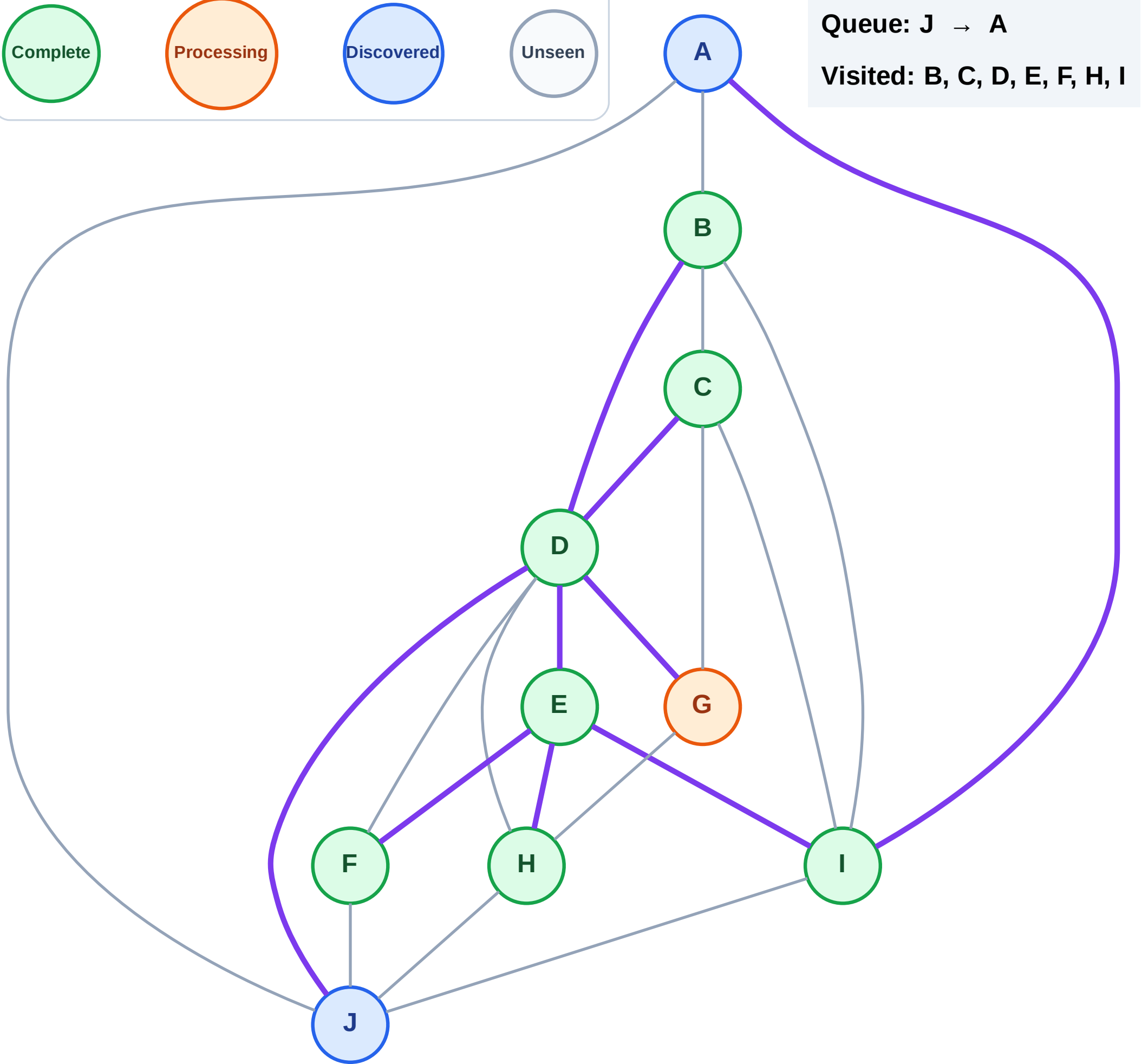
Processing

Discovered

Unseen

Queue: J → A

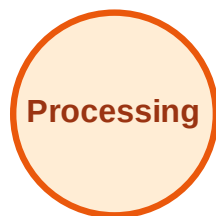
Visited: B, C, D, E, F, H, I



BFS Traversal

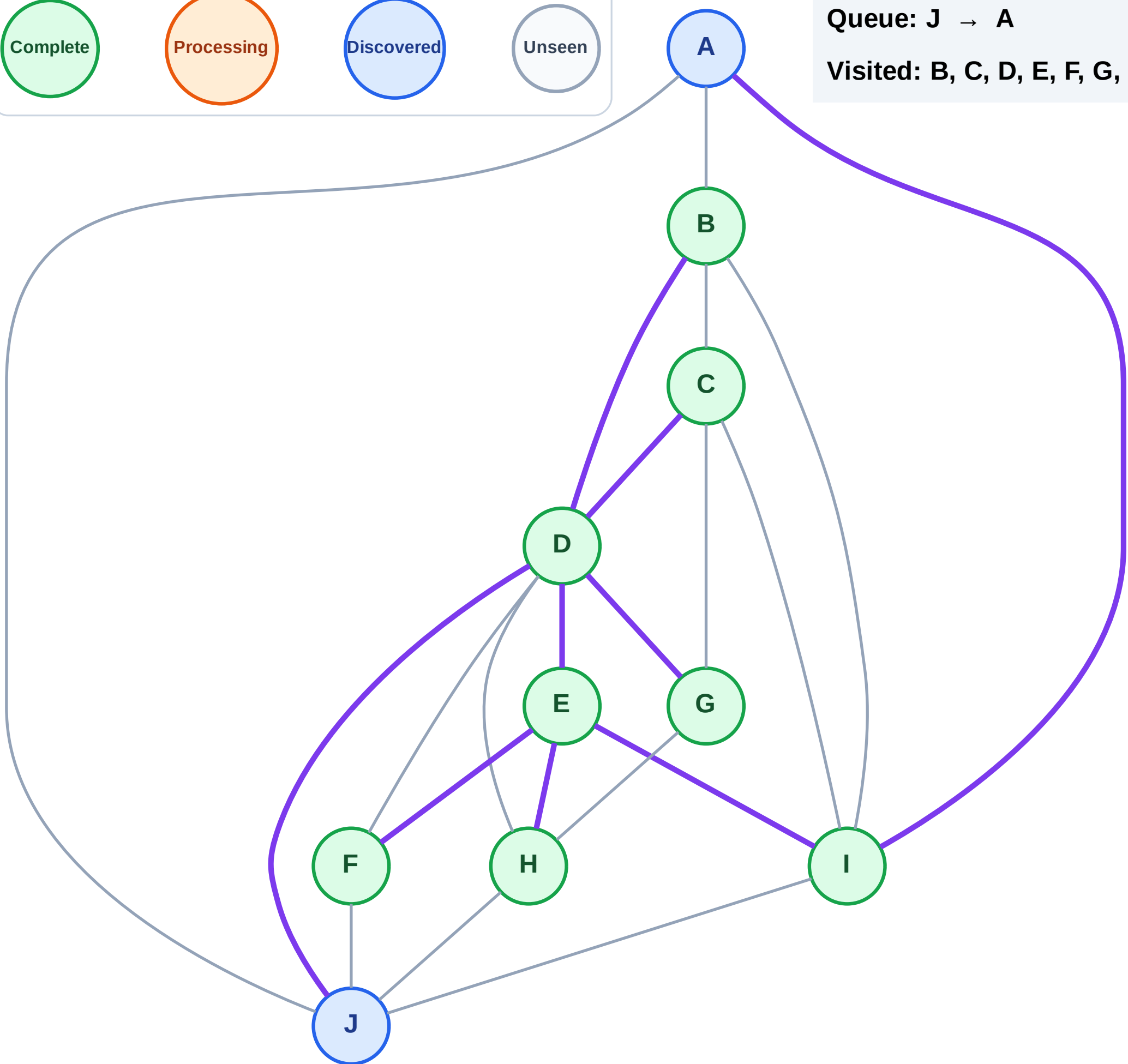
Step 17: No new neighbors from G

Legend



Queue: J → A

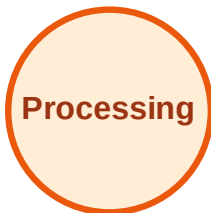
Visited: B, C, D, E, F, G, H, I



BFS Traversal

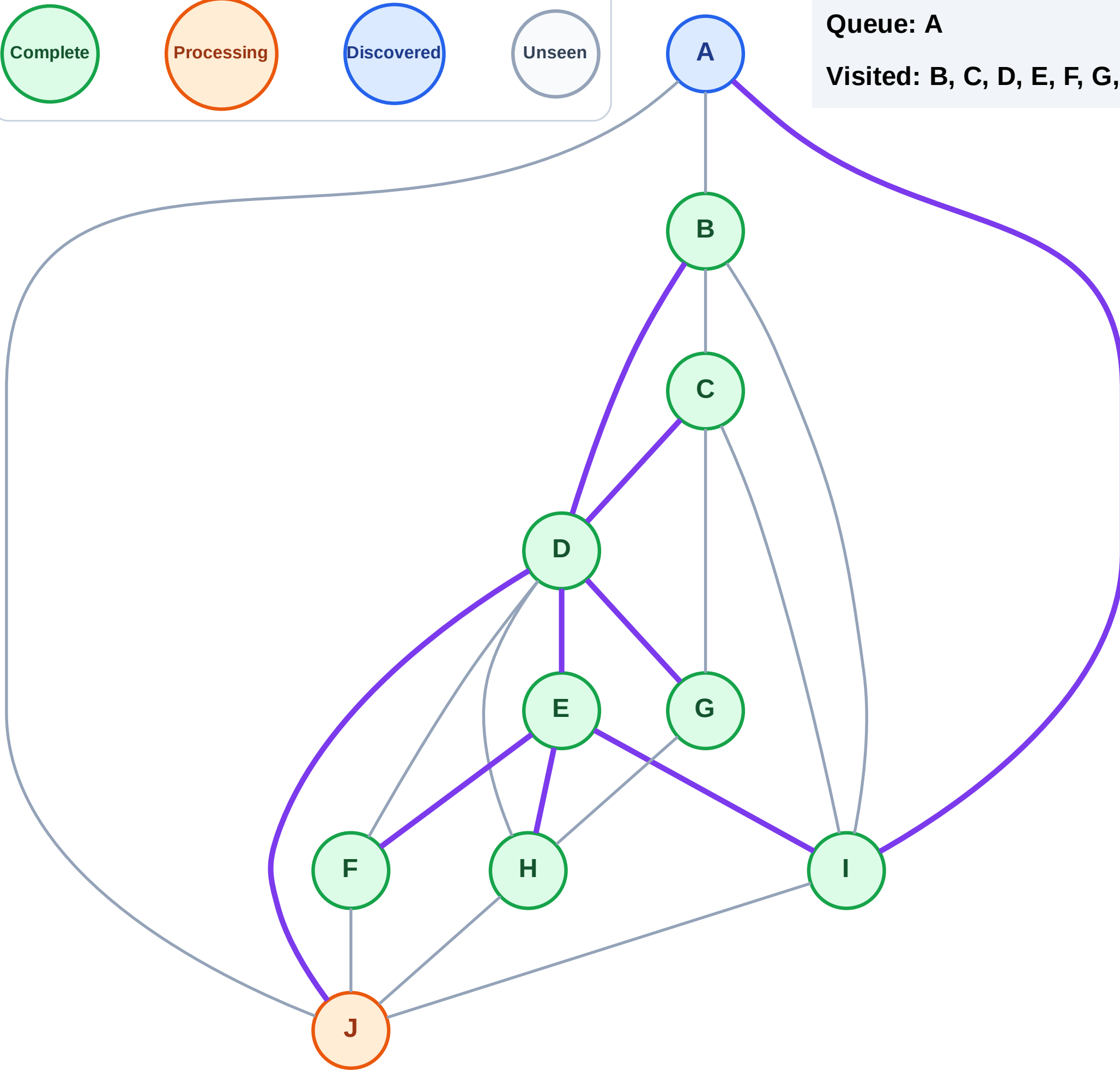
Step 18: Process node J

Legend



Queue: A

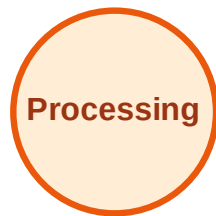
Visited: B, C, D, E, F, G, H, I



BFS Traversal

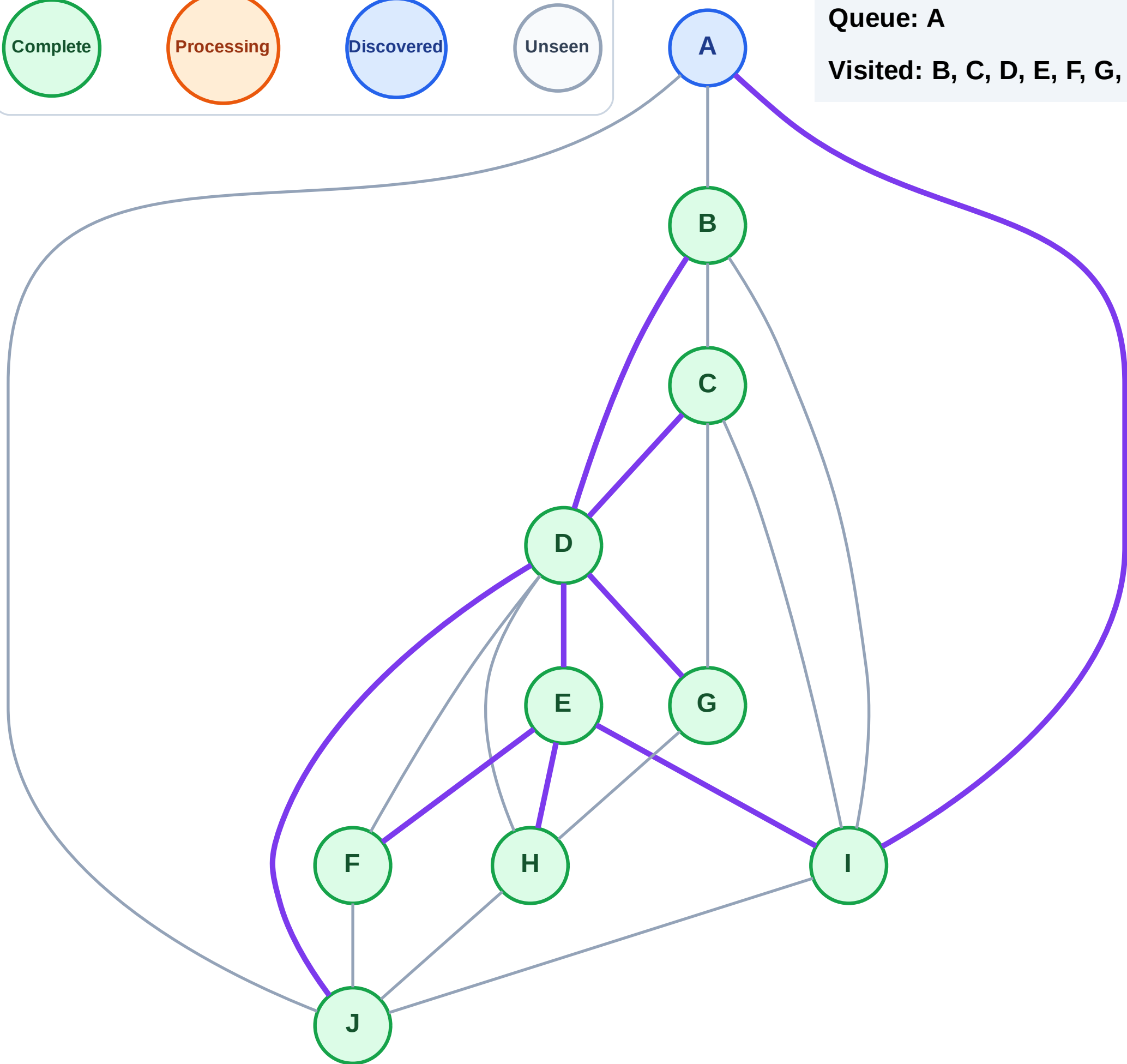
Step 19: No new neighbors from J

Legend



Queue: A

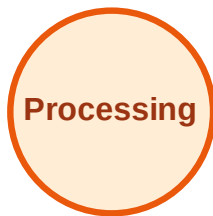
Visited: B, C, D, E, F, G, H, I, J



BFS Traversal

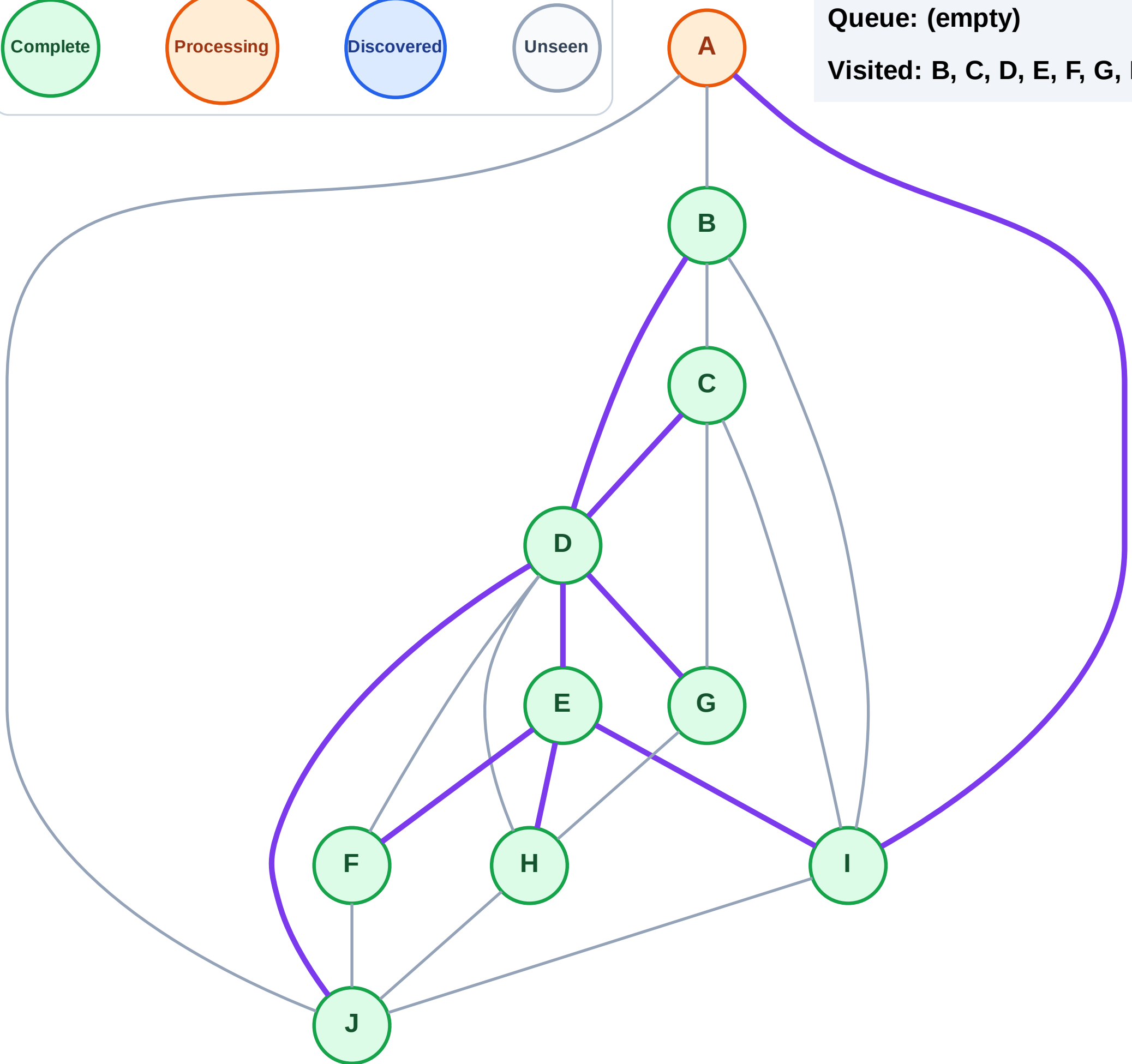
Step 20: Process node A

Legend



Queue: (empty)

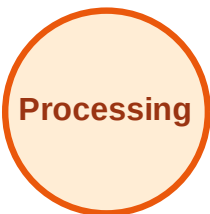
Visited: B, C, D, E, F, G, H, I, J



BFS Traversal

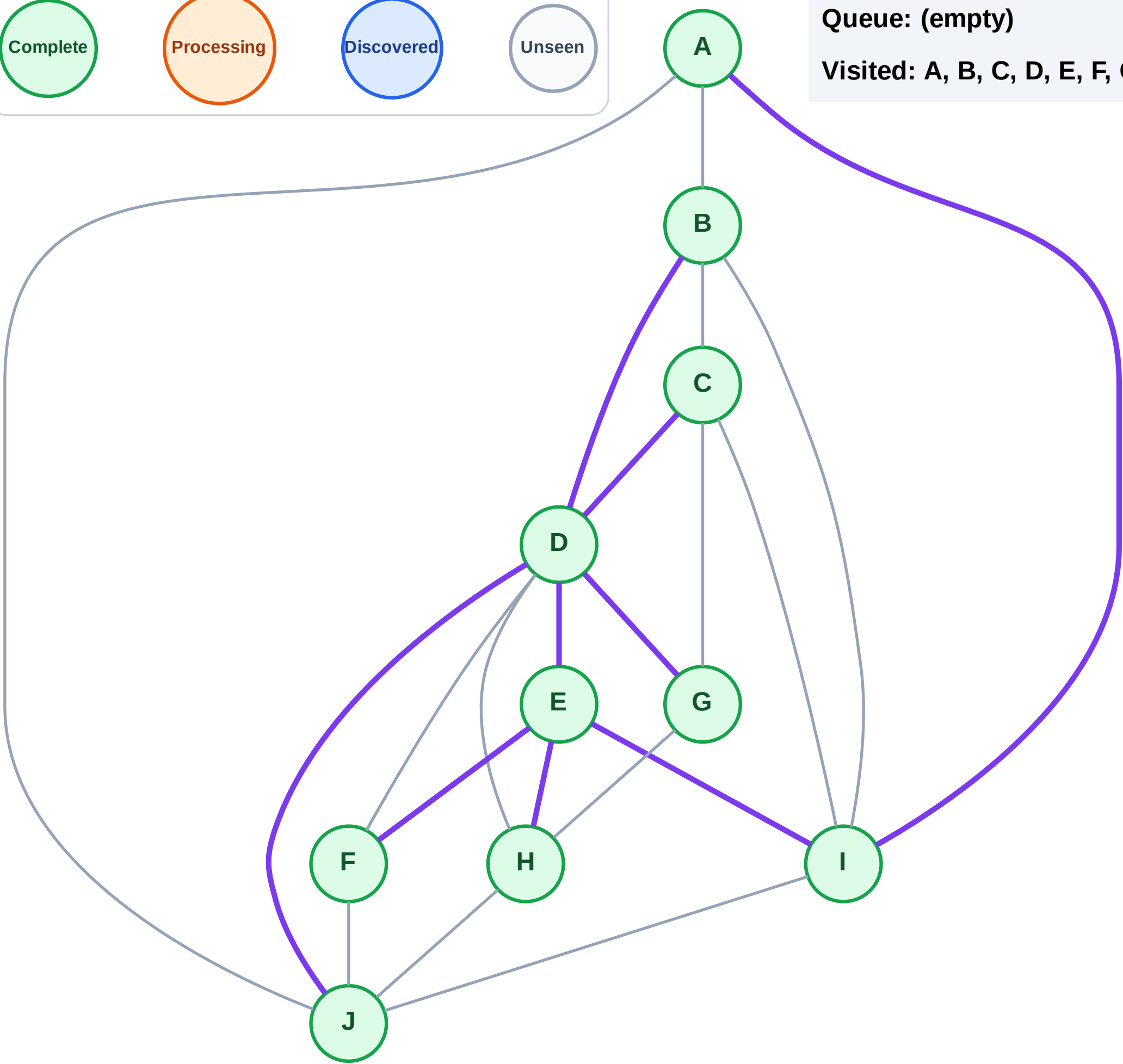
Step 21: No new neighbors from A

Legend



Queue: (empty)

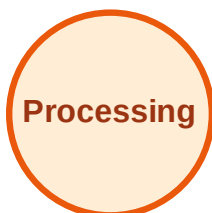
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

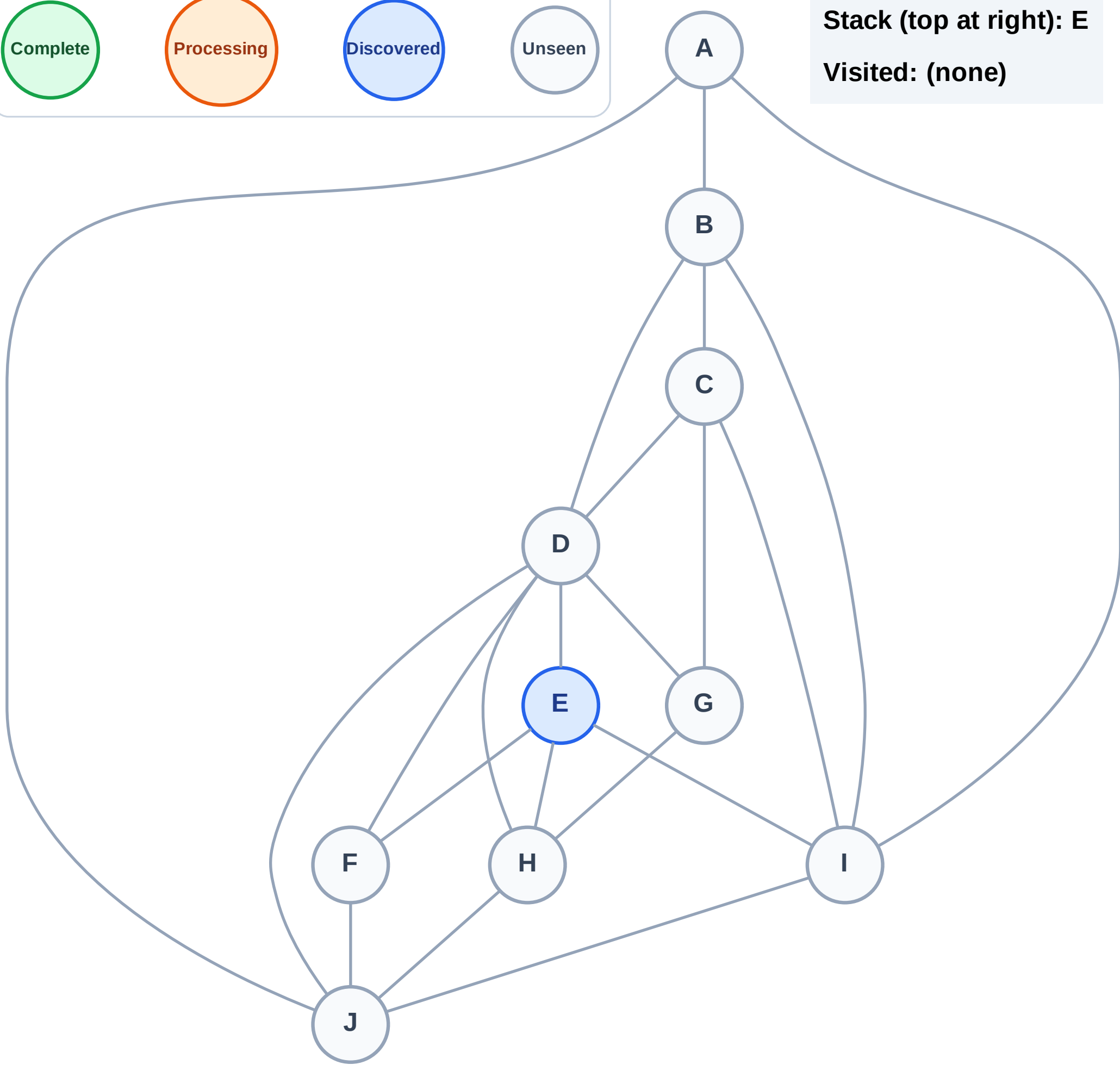
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

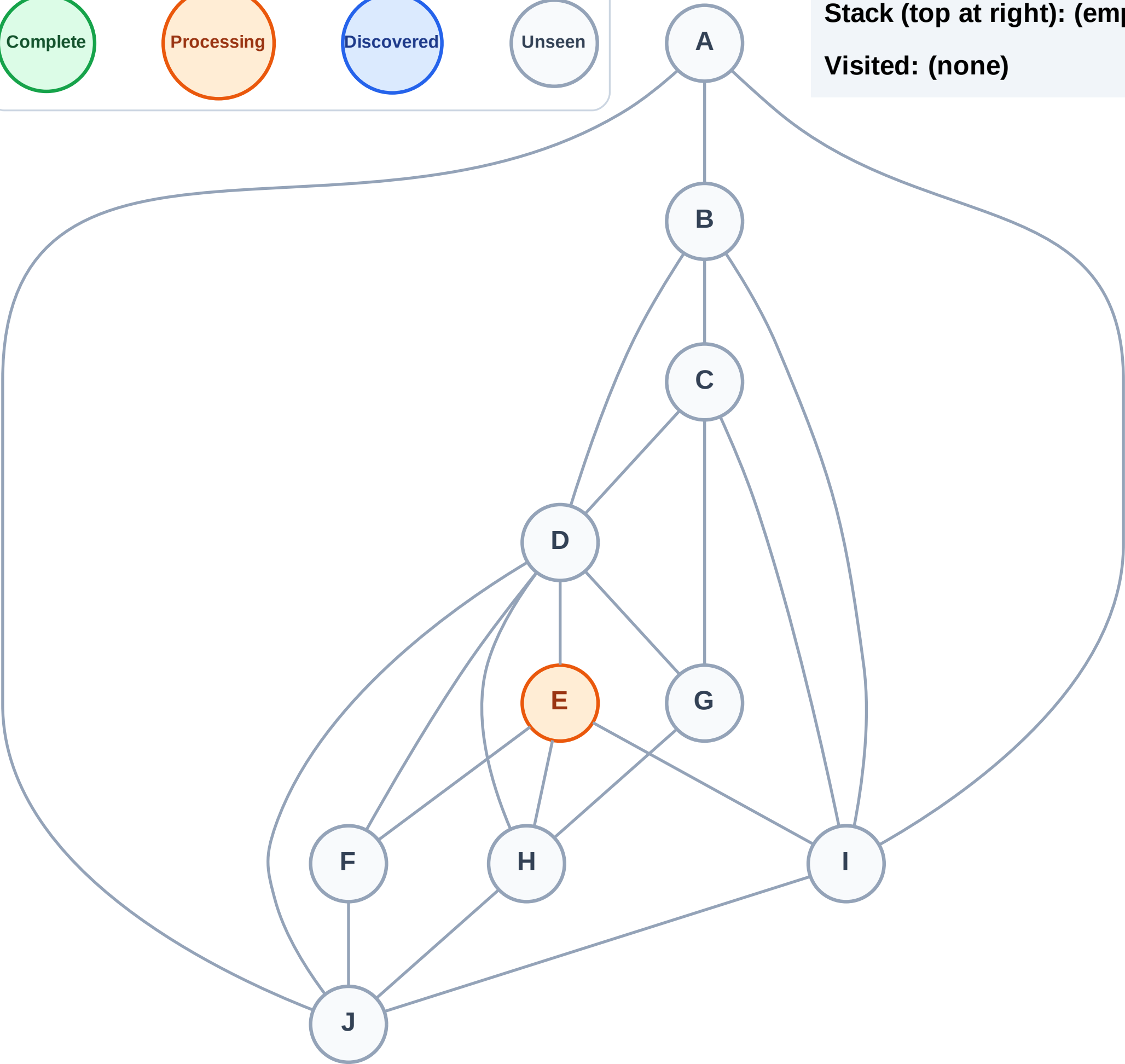
Complete

Processing

Discovered

Unseen

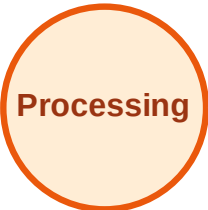
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

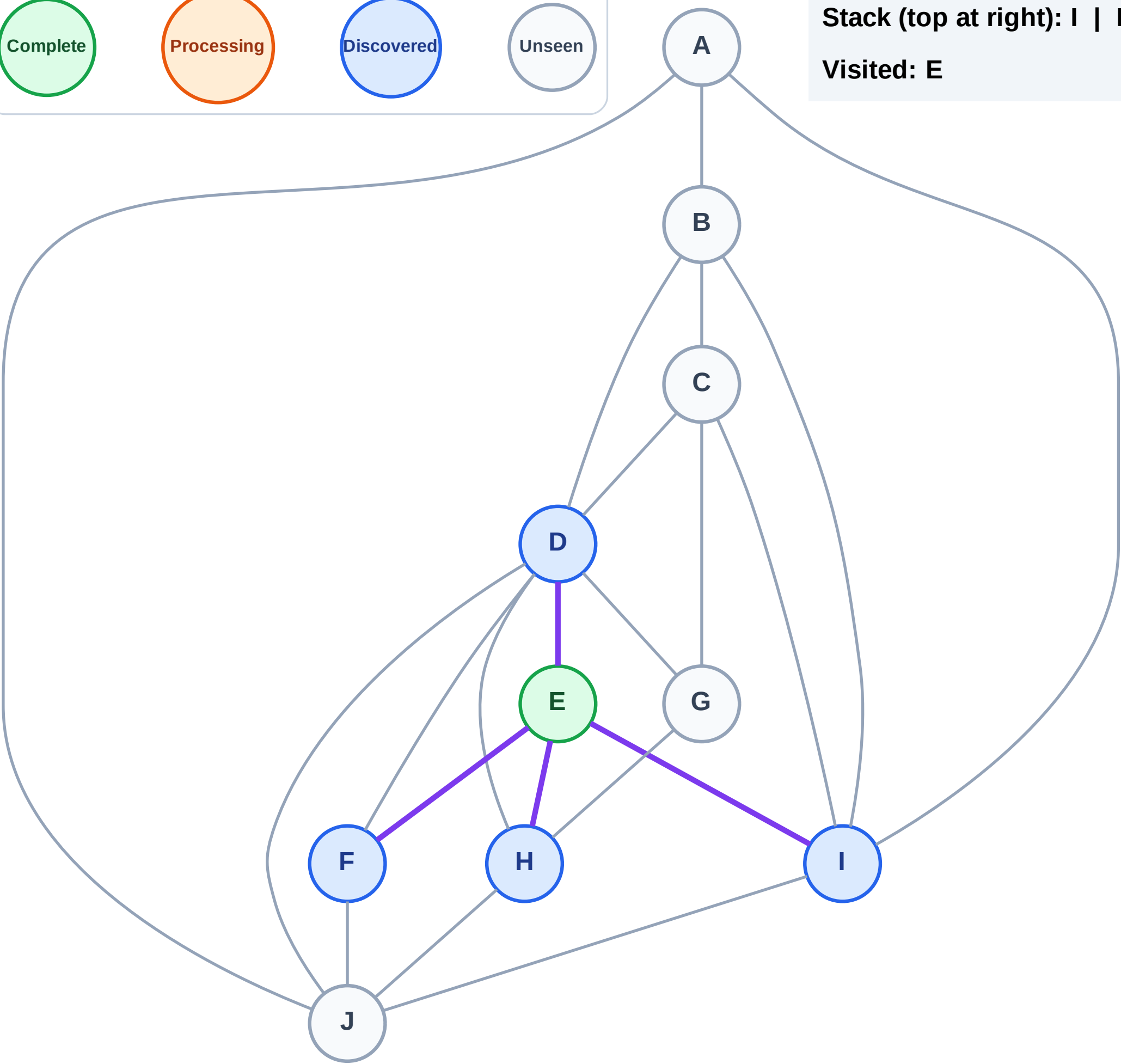
Step 3: Discover I, H, F, D from E

Legend



Stack (top at right): I | H | F | D

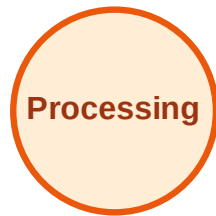
Visited: E



DFS Traversal

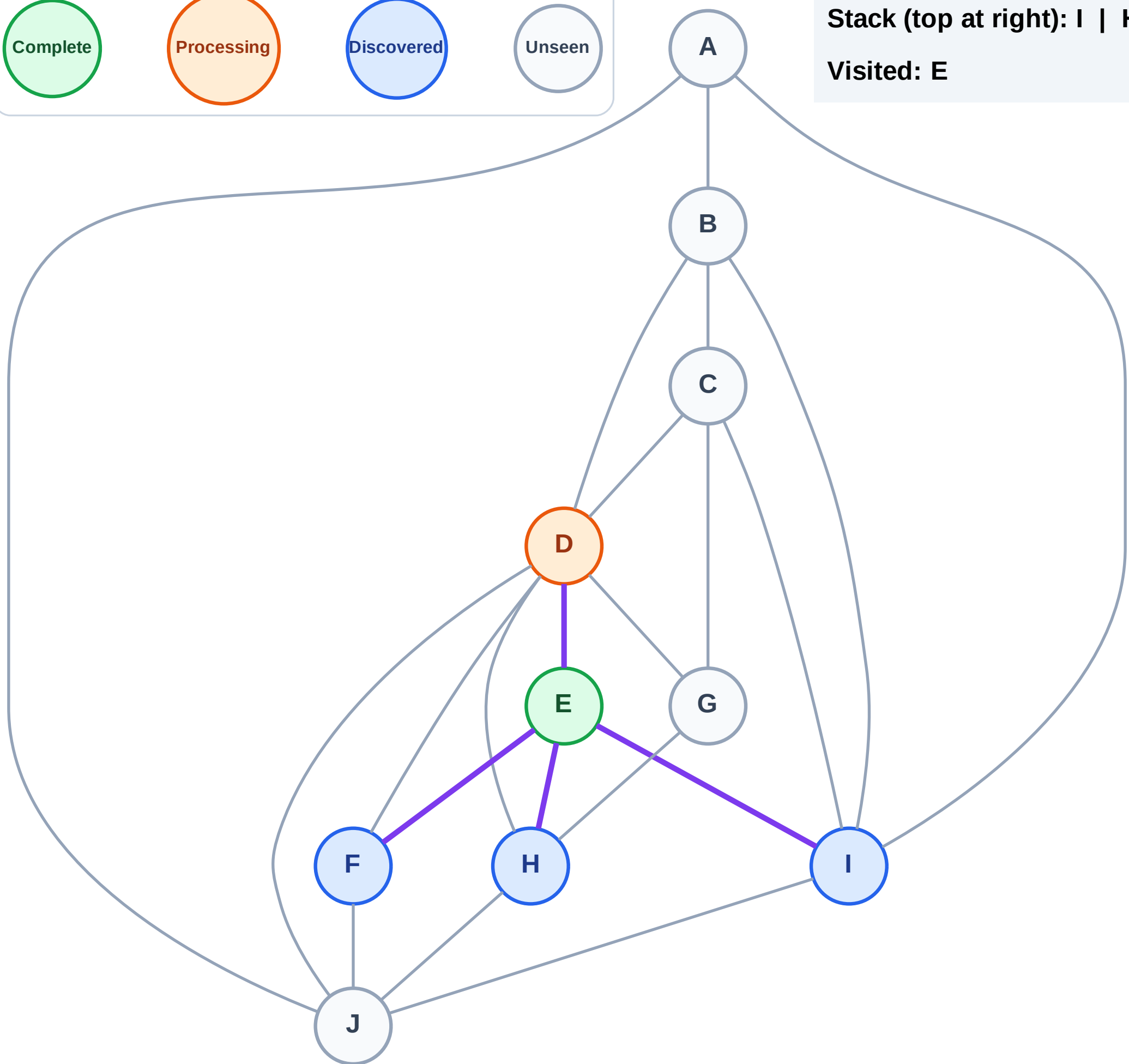
Step 4: Process node D

Legend



Stack (top at right): I | H | F

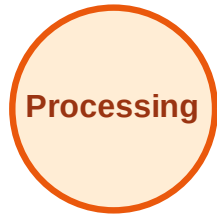
Visited: E



DFS Traversal

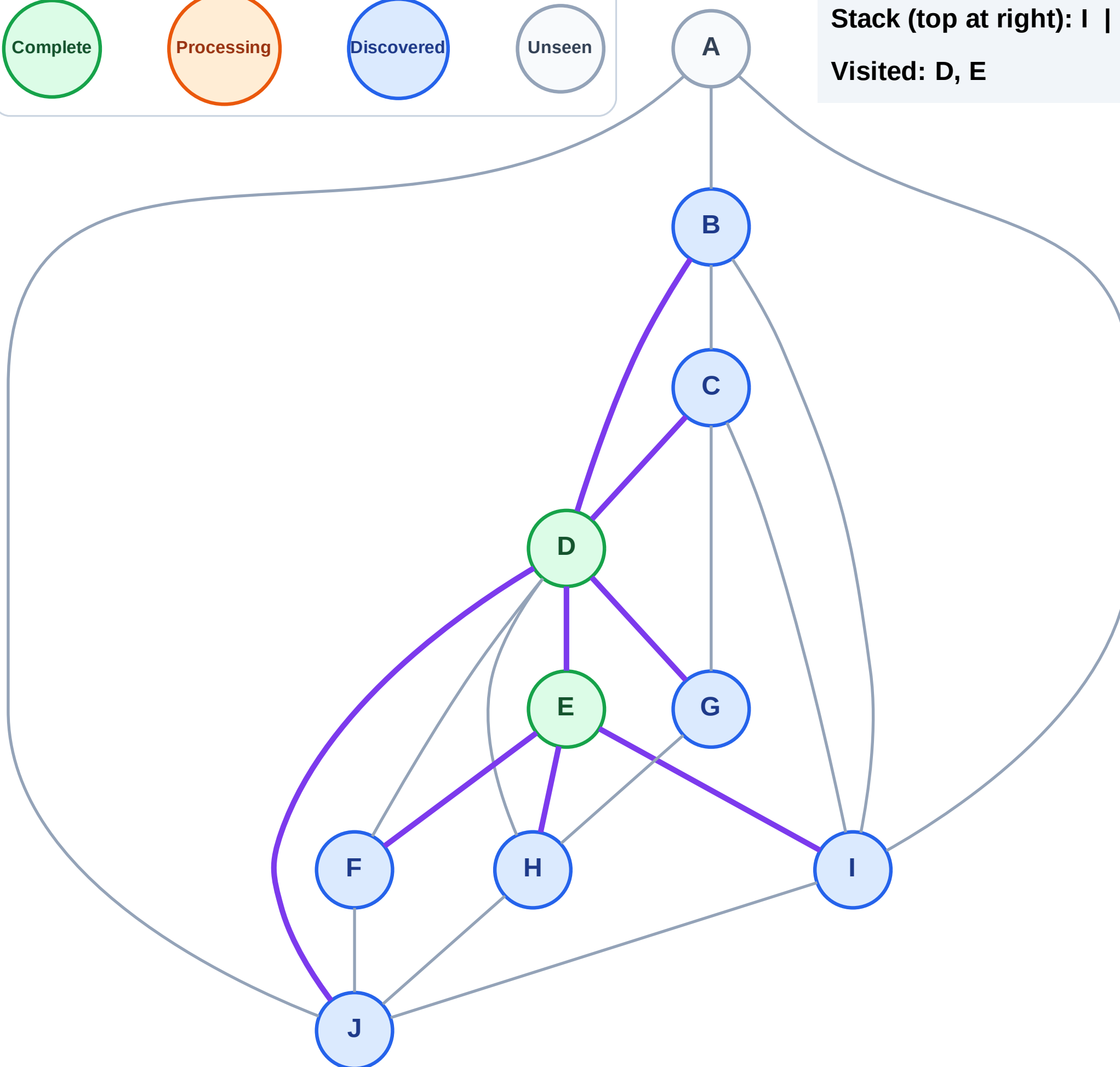
Step 5: Discover J, G, C, B from D

Legend

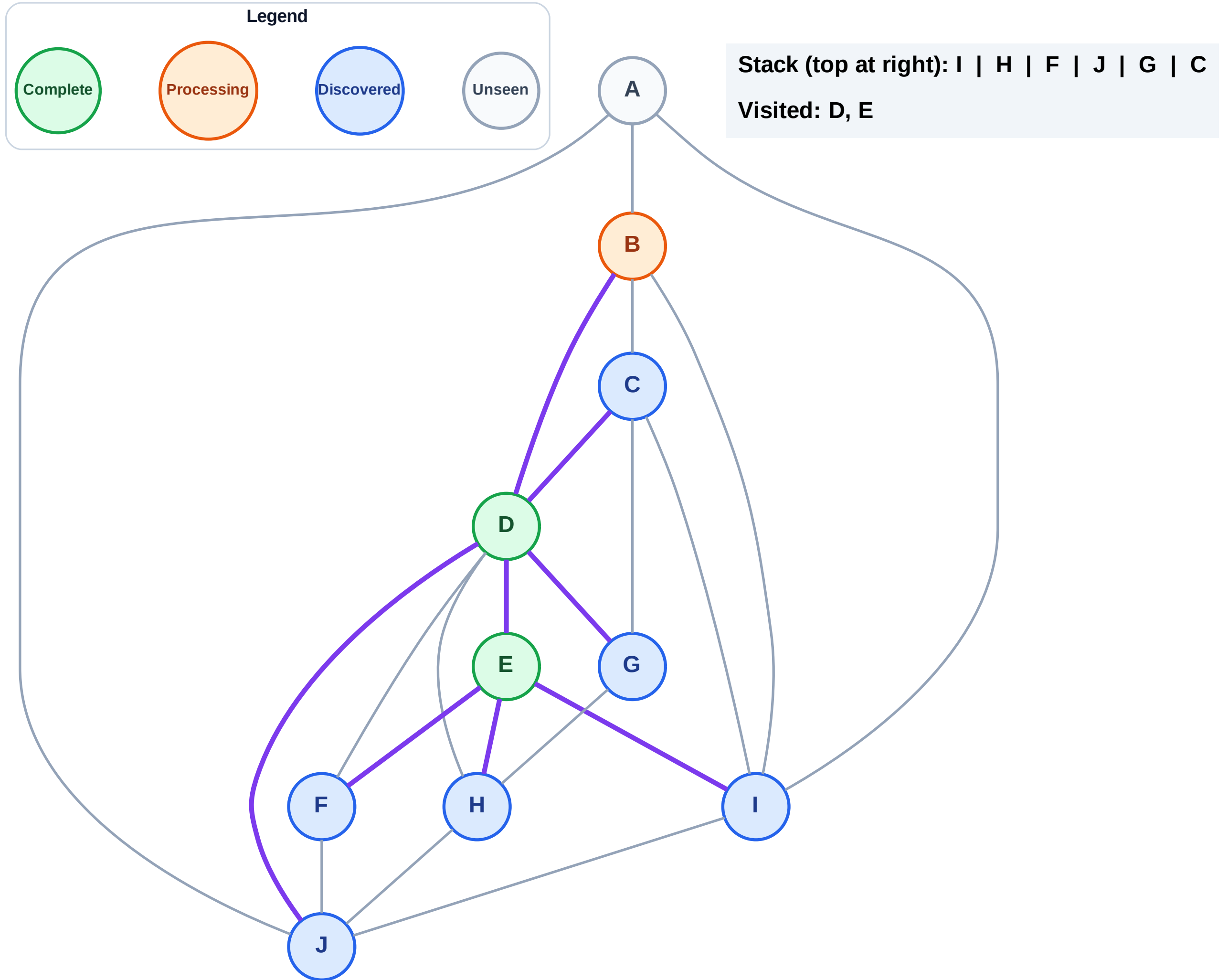


Stack (top at right): I | H | F | J | G | C | B

Visited: D, E



Step 6: Process node B



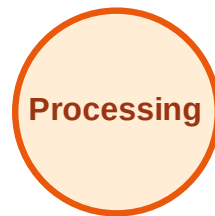
Step 7: Discover A from B



DFS Traversal

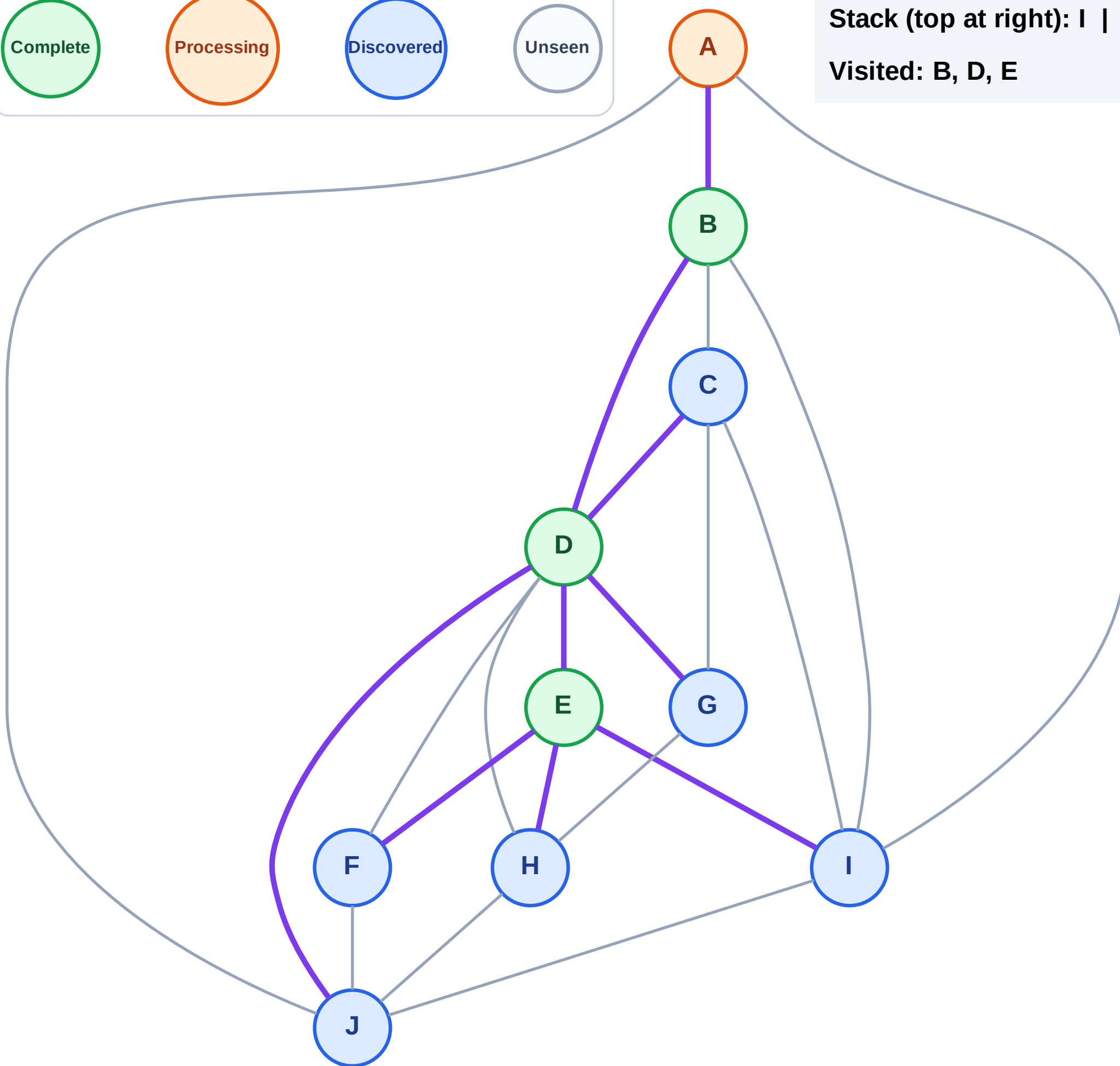
Step 8: Process node A

Legend



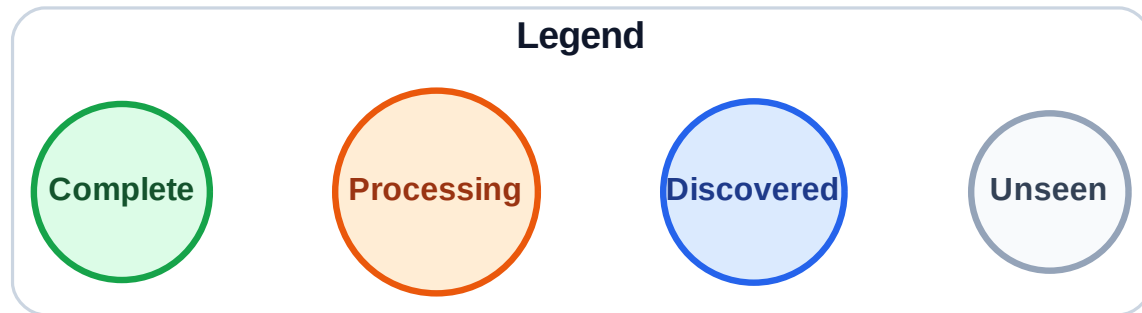
Stack (top at right): I | H | F | J | G | C

Visited: B, D, E



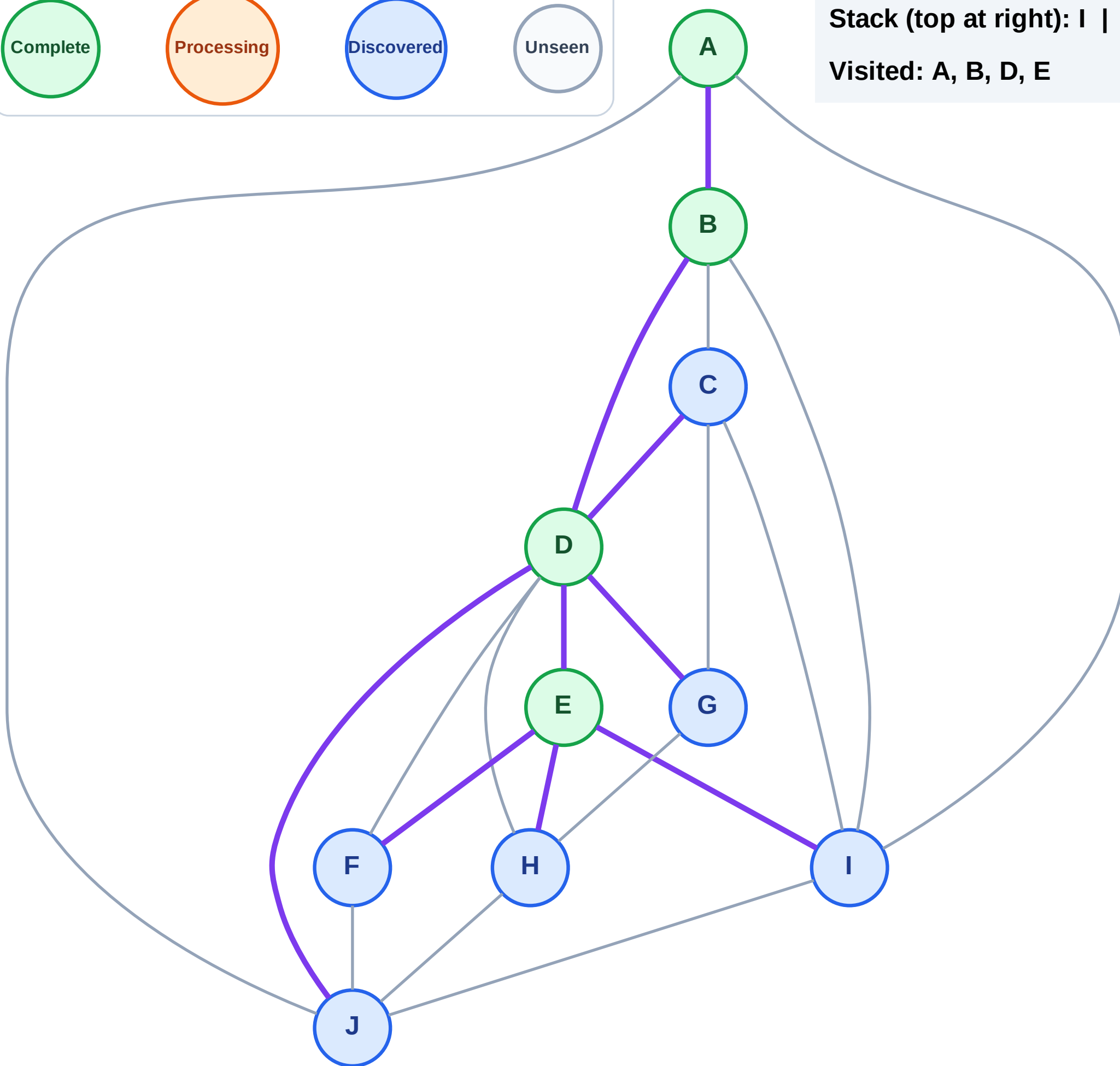
Step 9: No new neighbors from A

Step 9: No new neighbors from A



Stack (top at right): I | H | F | J | G | C

Visited: A, B, D, E



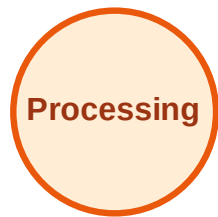
DFS Traversal

Step 10: Process node C

Legend



Complete



Processing



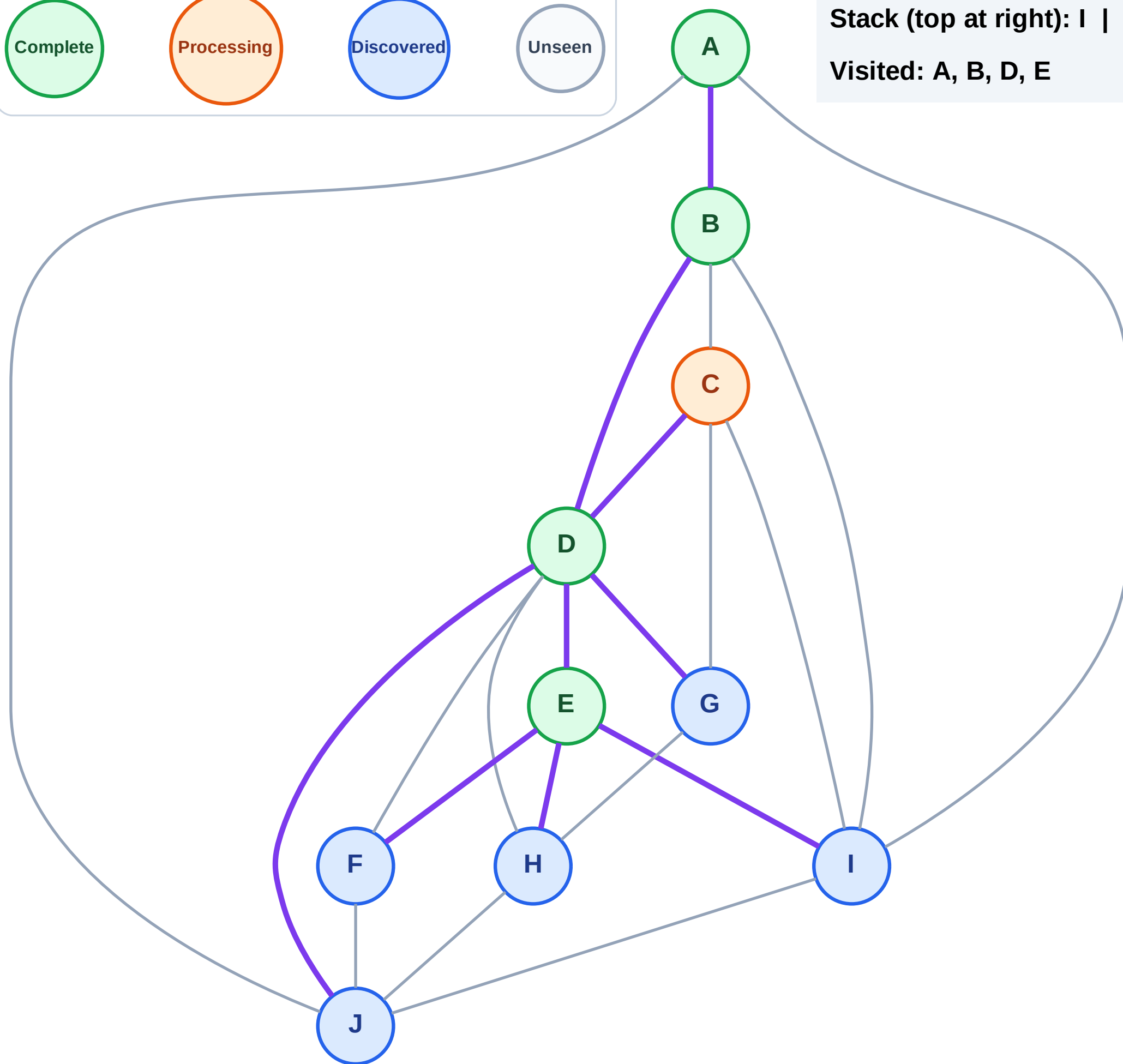
Discovered



Unseen

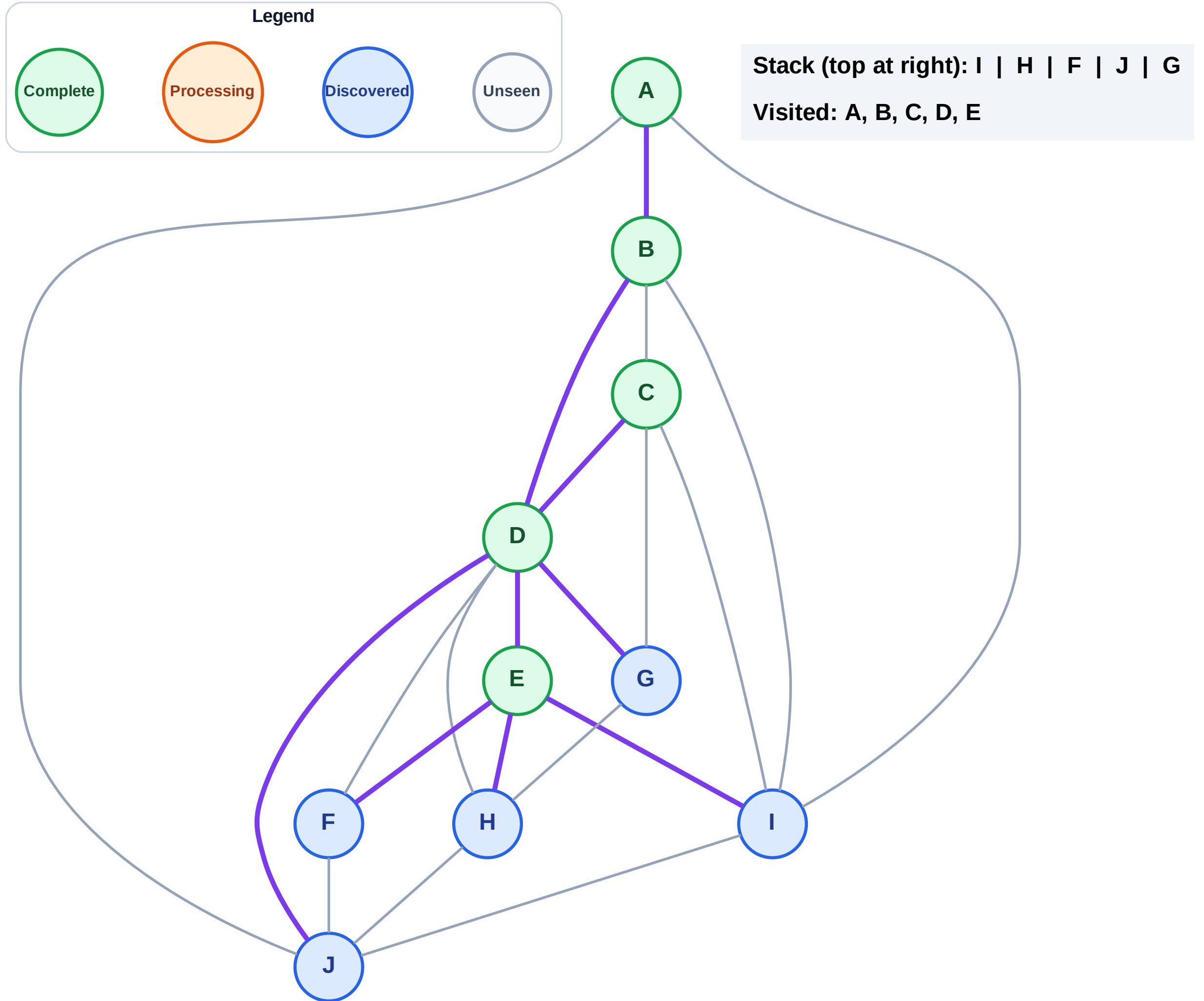
Stack (top at right): I | H | F | J | G

Visited: A, B, D, E



Step 11: No new neighbors from C

Step 11: No new neighbors from C



DFS Traversal

Step 12: Process node G

Legend



Complete



Processing



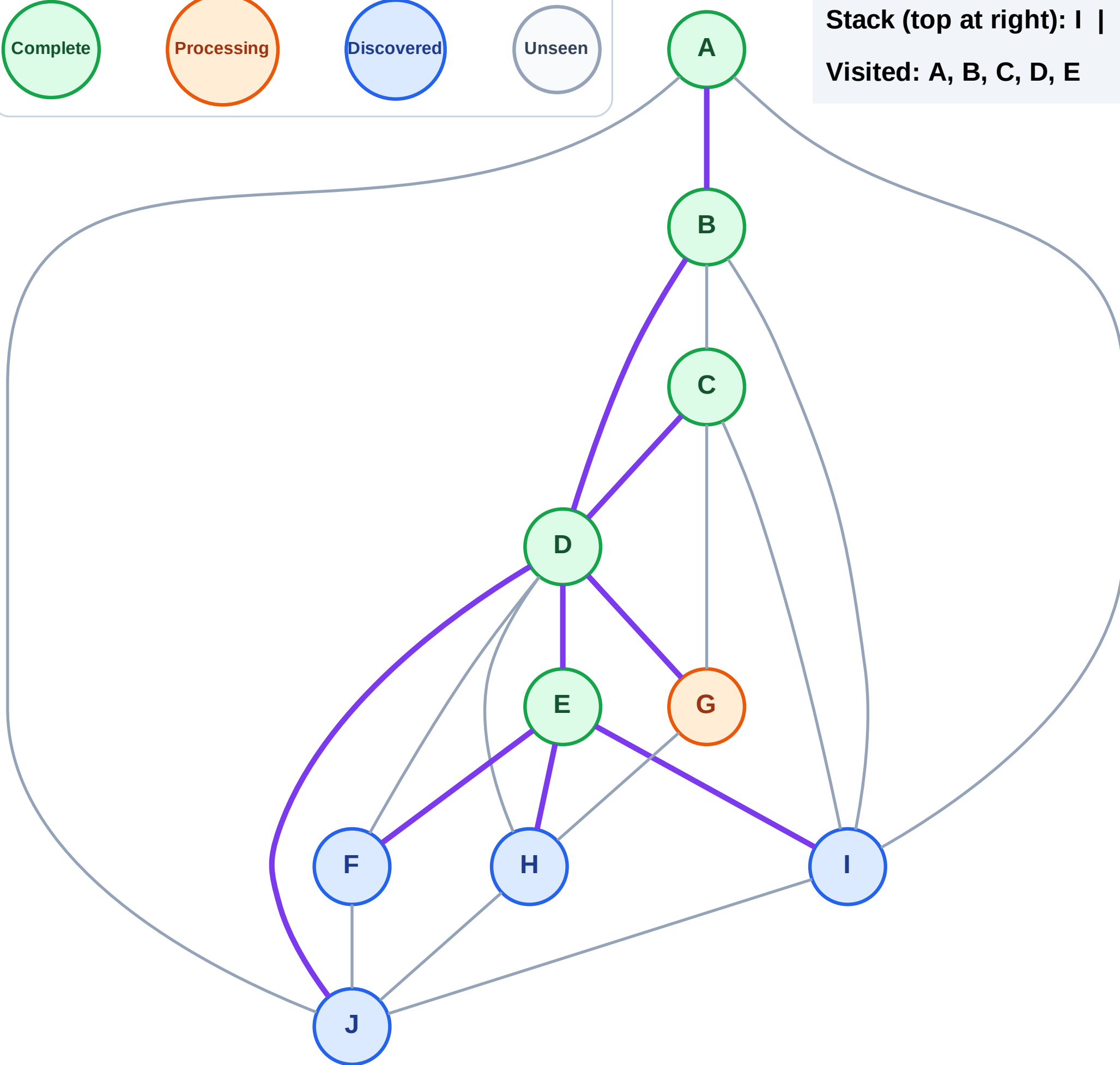
Discovered



Unseen

Stack (top at right): I | H | F | J

Visited: A, B, C, D, E



DFS Traversal

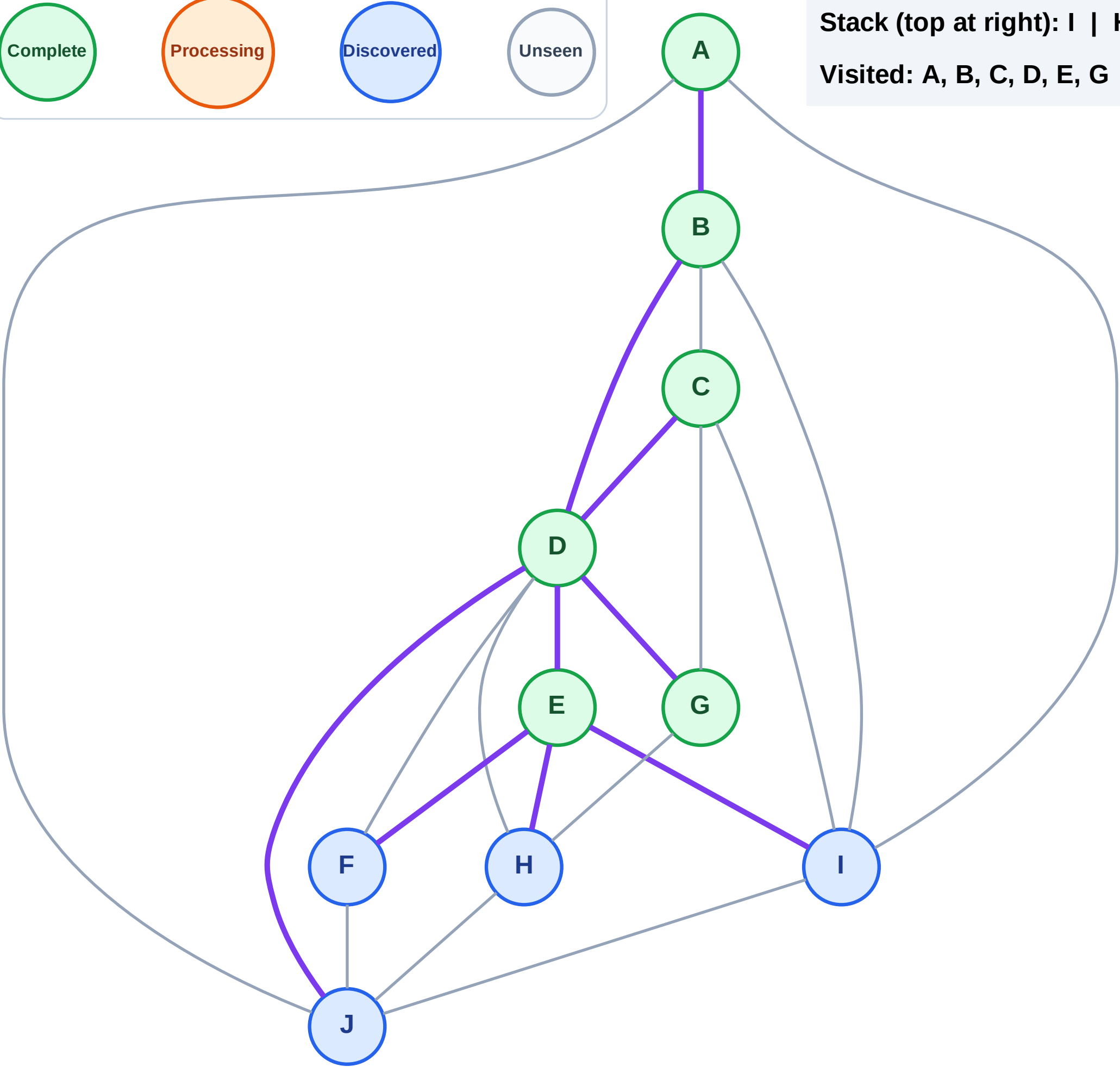
Step 13: No new neighbors from G

Legend



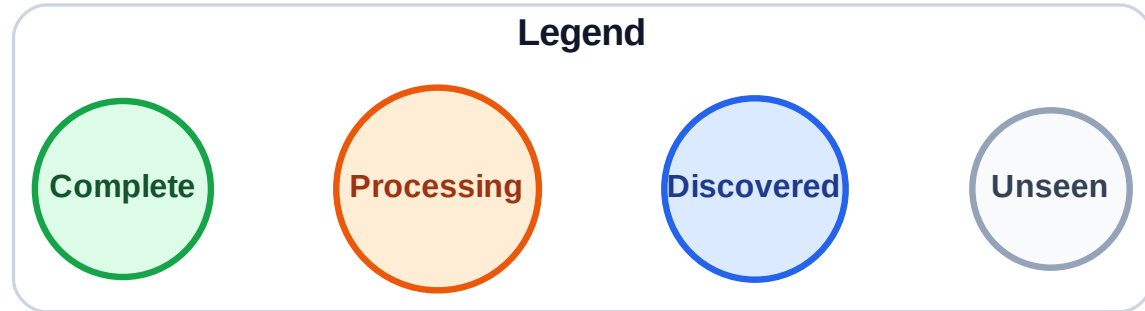
Stack (top at right): I | H | F | J

Visited: A, B, C, D, E, G

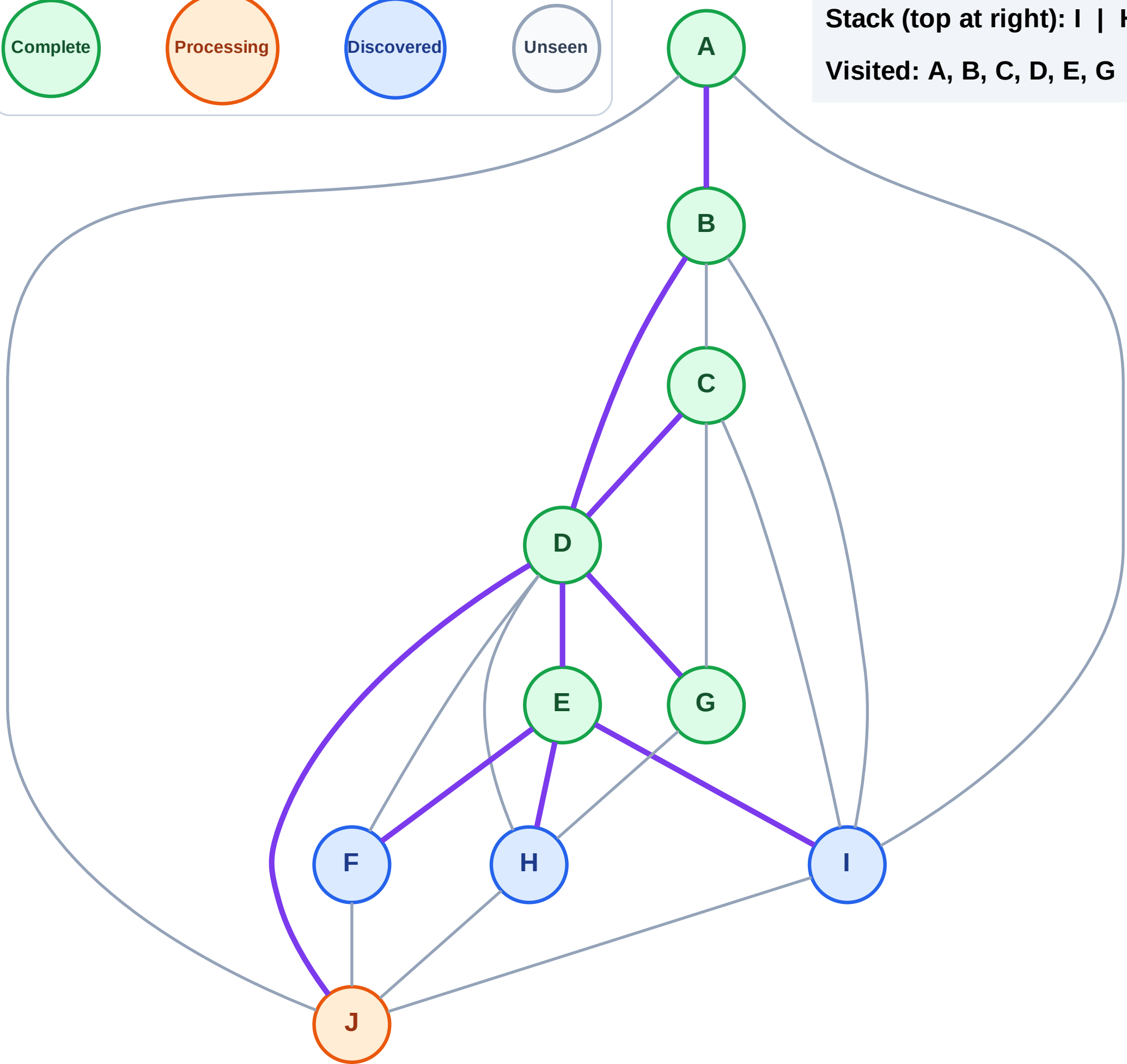


Step 14: Process node J

Page 10 of 10



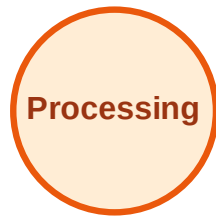
Visited: A, B, C, D, E, G



DFS Traversal

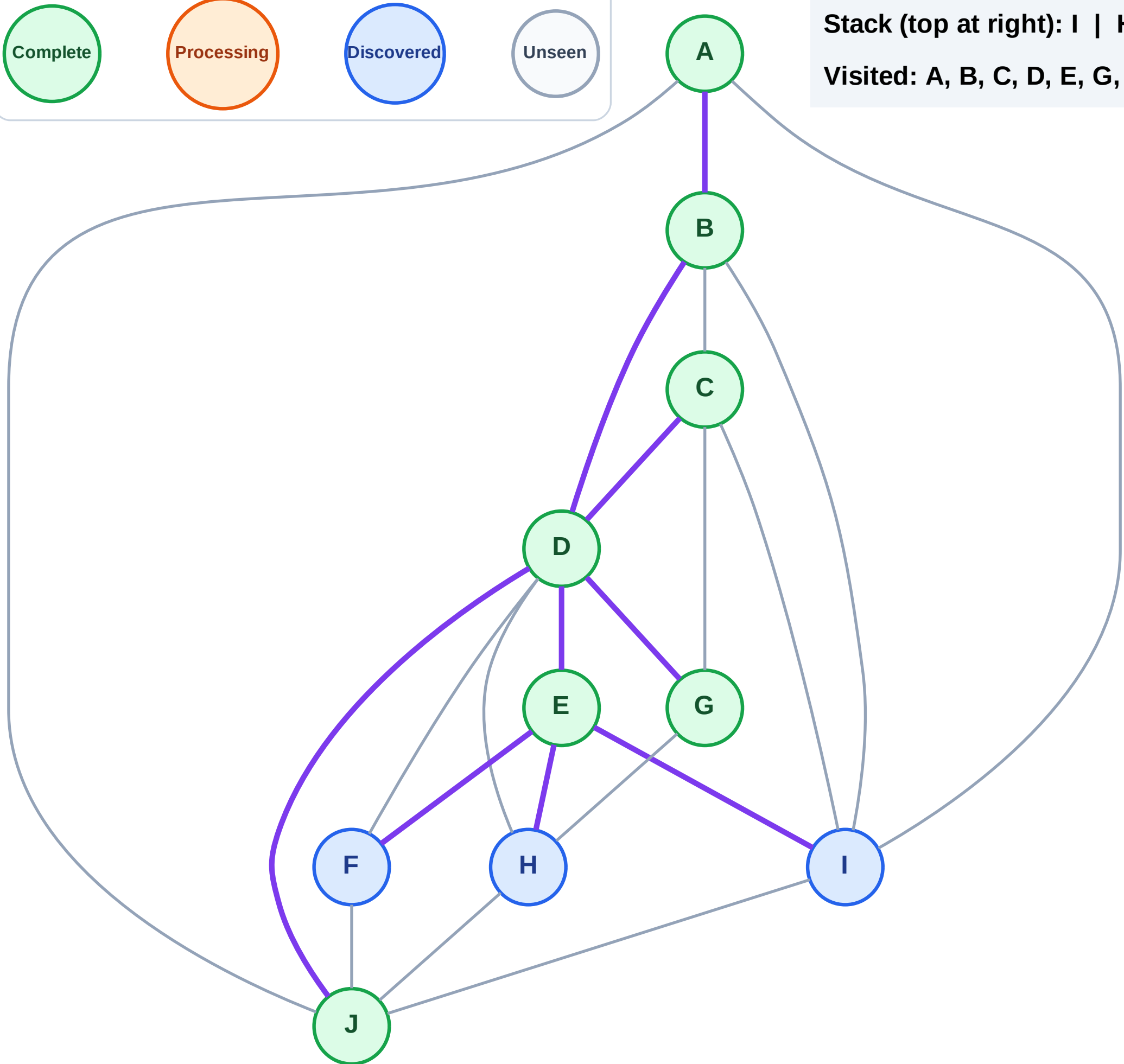
Step 15: No new neighbors from J

Legend



Stack (top at right): I | H | F

Visited: A, B, C, D, E, G, J



DFS Traversal

Step 16: Process node F

Legend

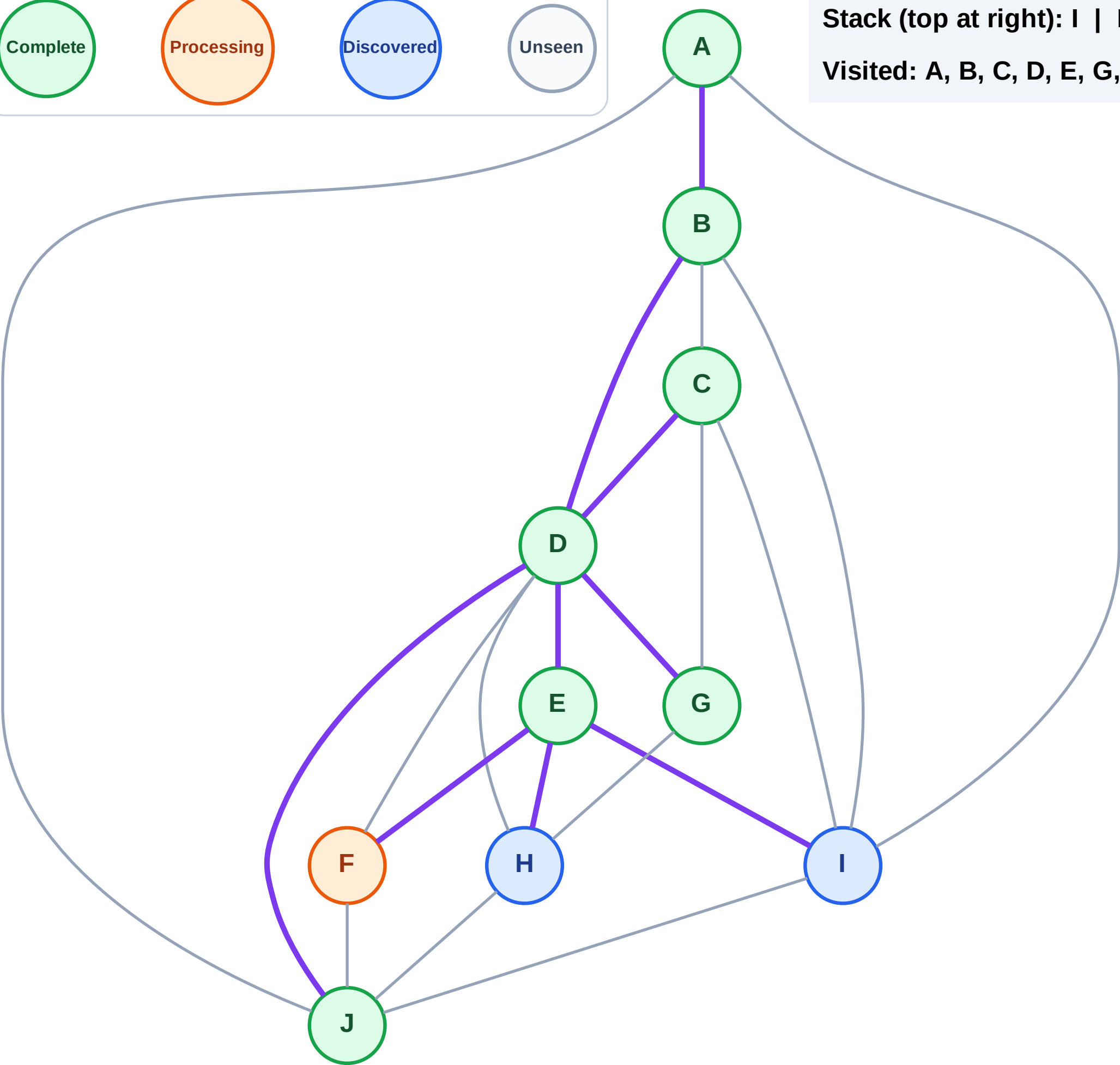
Complete

Processing

Discovered

Unseen

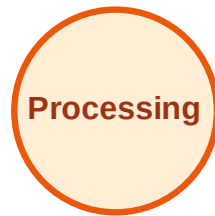
Stack (top at right): I | H
Visited: A, B, C, D, E, G, J



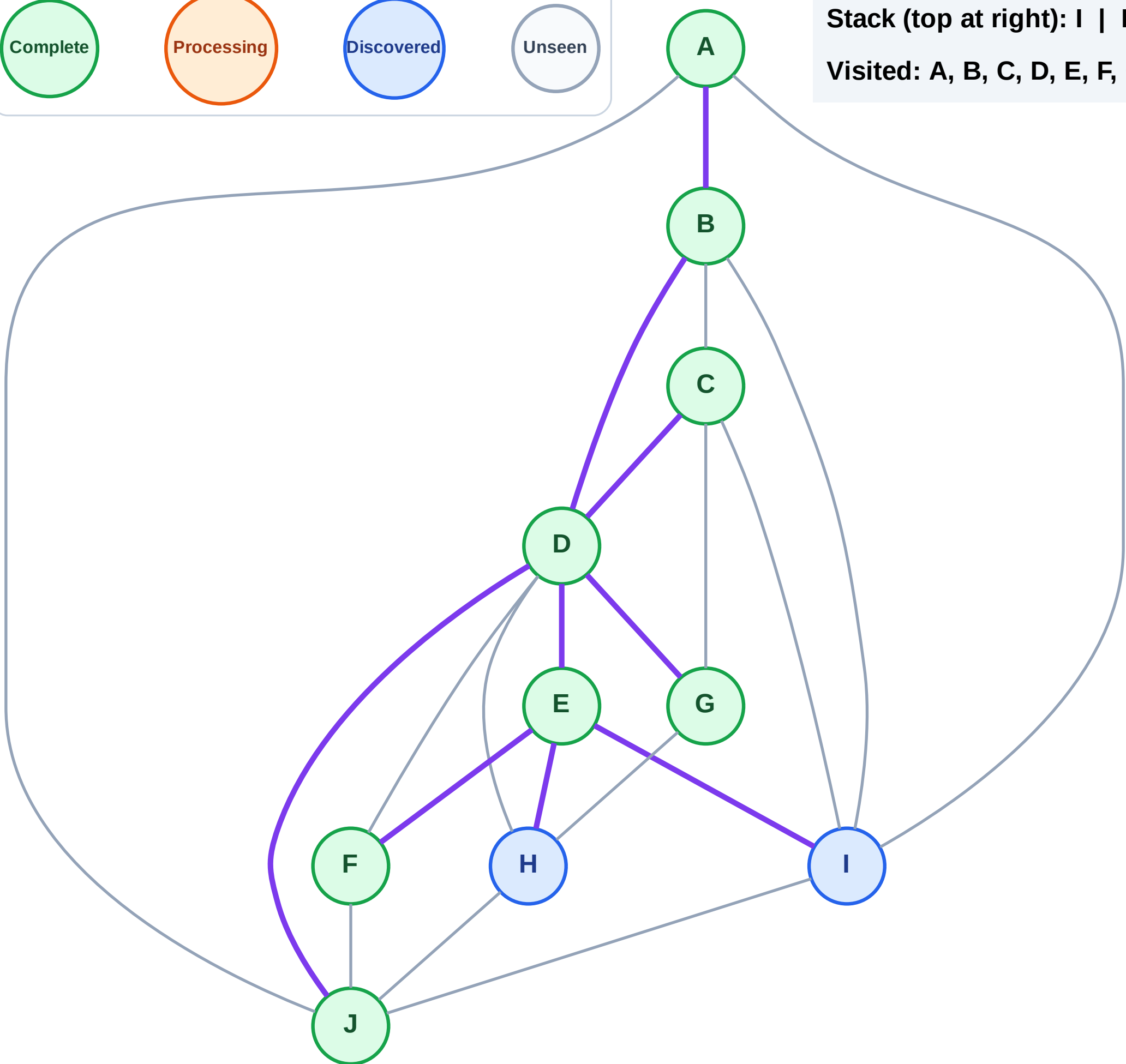
DFS Traversal

Step 17: No new neighbors from F

Legend



Stack (top at right): I | H
Visited: A, B, C, D, E, F, G, J



DFS Traversal

Step 18: Process node H

Legend

Complete

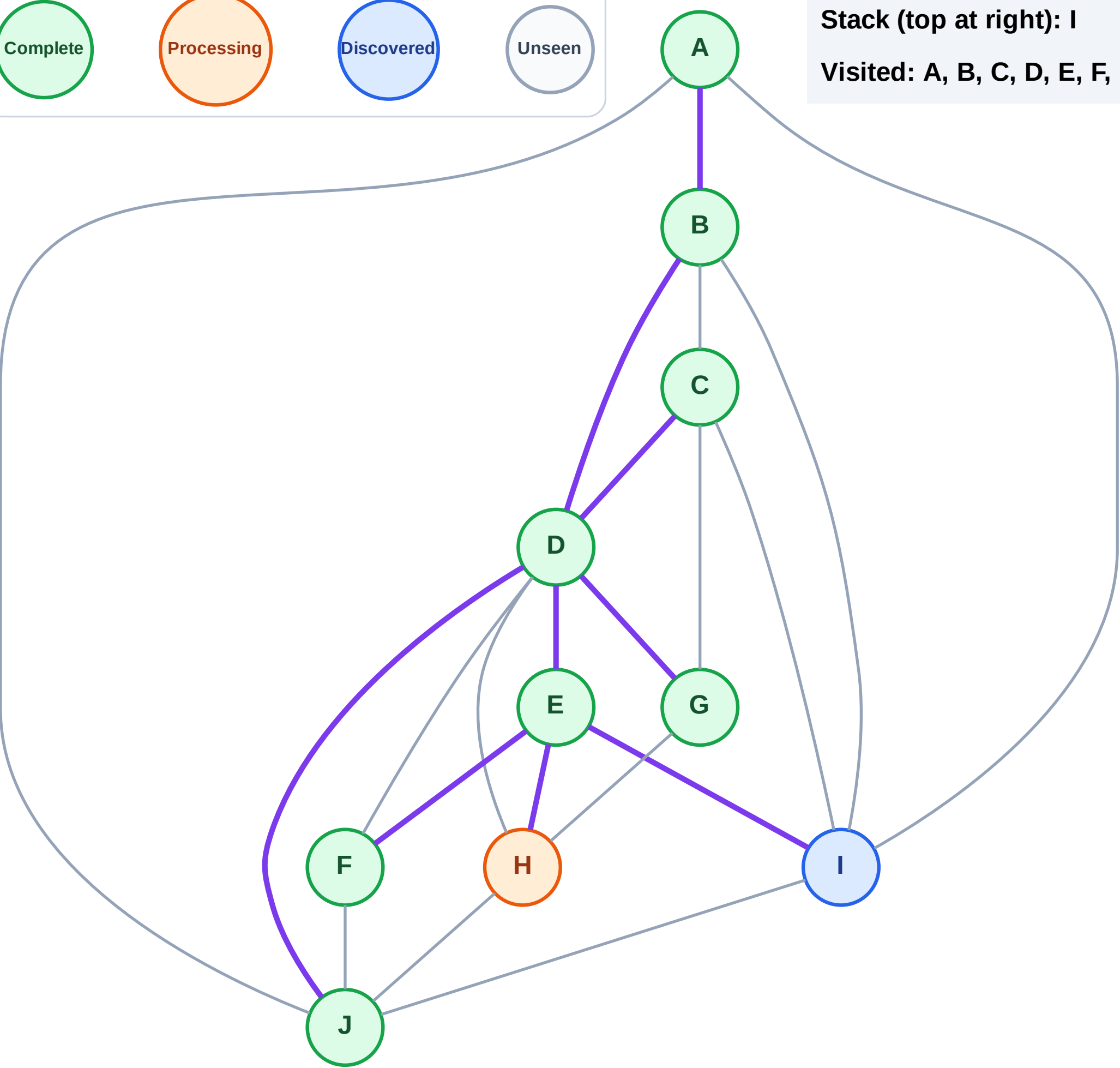
Processing

Discovered

Unseen

Stack (top at right): I

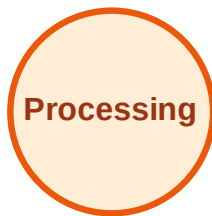
Visited: A, B, C, D, E, F, G, J



DFS Traversal

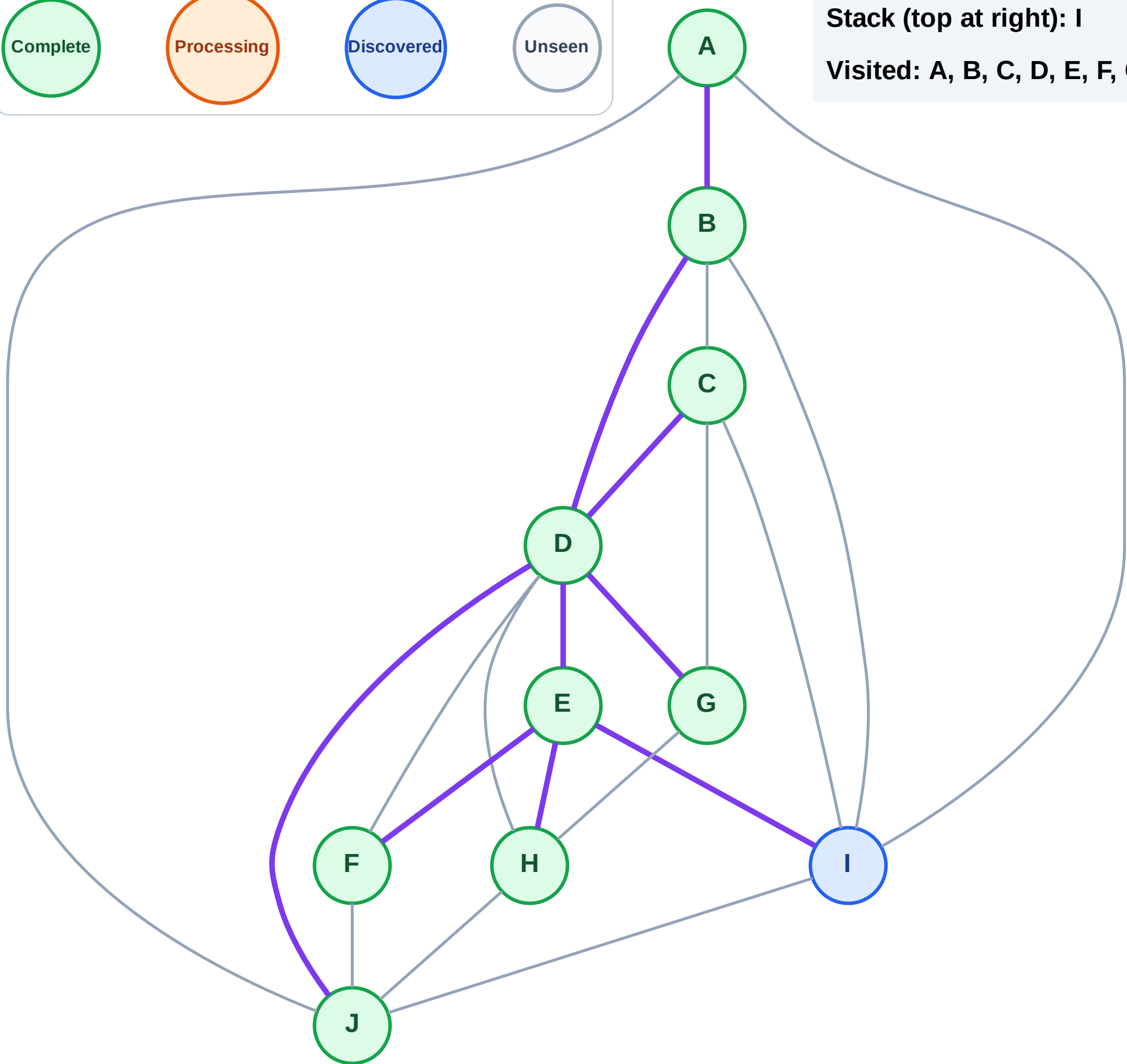
Step 19: No new neighbors from H

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

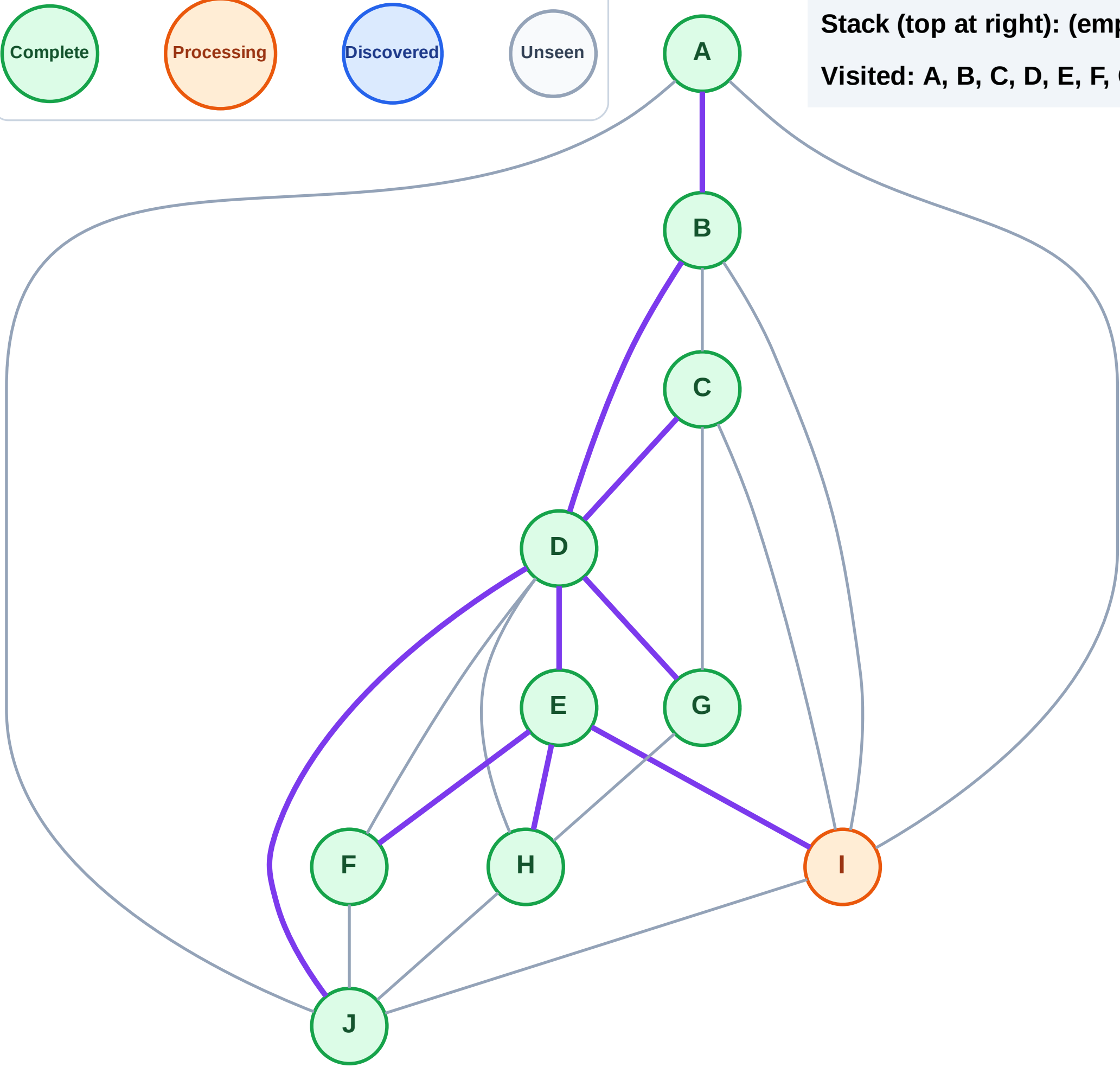
Step 20: Process node I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

